PART II

Permutations, Cosets, and Direct Products

Section 8 Groups of Permutations
Section 9 Orbits, Cycles, and the Alternating Groups
Section 10 Cosets and the Theorem of Lagrange
Section 11 Direct Products and Finitely Generated Abelian Groups
Section 12 †Plane Isometries

SECTION 8 GROUPS OF PERMUTATIONS

We have seen examples of groups of numbers, like the groups $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ under addition. We have also introduced groups of matrices, like the group $GL(2, \mathbb{R})$. Each element A of $GL(2, \mathbb{R})$ yields a transformation of the plane $\mathbb{R}^2$ into itself; namely, if we regard $\mathbf{x}$ as a 2-component column vector, then $A\mathbf{x}$ is also a 2-component column vector. The group $GL(2, \mathbb{R})$ is typical of many of the most useful groups in that its elements *act on things* to transform them. Often, an action produced by a group element can be regarded as a *function*, and the binary operation of the group can be regarded as *function composition*. In this section, we construct some finite groups whose elements, called *permutations*, act on finite sets. These groups will provide us with examples of finite nonabelian groups. We shall show that any finite group is structurally the same as some group of permutations. Unfortunately, this result, which sounds very powerful, does not turn out to be particularly useful to us.

You may be familiar with the notion of a permutation of a set as a rearrangement of the elements of the set. Thus for the set $\{1, 2, 3, 4, 5\}$, a rearrangement of the elements could be given schematically as in Fig. 8.1, resulting in the new arrangement $\{4, 2, 5, 3, 1\}$. Let us think of this schematic diagram in Fig. 8.1 as a function mapping of each element listed in the left column into a single (not necessarily different) element from the same set listed at the right. Thus 1 is carried into 4, 2 is mapped into 2, and so on. Furthermore, to be a permutation of the set, this mapping must be such that each element appears in the right column once and only once. For example, the diagram in Fig. 8.2 does *not* give a permutation, for 3 appears twice while 1 does not appear at all in the right column. We now define a permutation to be such a mapping.

† Section 12 is not used in the remainder of the text.

1→4	1→3
2→2	2→2
3→5	3→4
4→3	4→5
5→1	5→3

8.1 Figure **8.2 Figure**

8.3 Definition A **permutation of a set** A is a function $\phi : A \to A$ that is both one to one and onto. ■

Permutation Groups

We now show that function composition $\circ$ is a binary operation on the collection of all permutations of a set A. We call this operation *permutation multiplication*. Let A be a set, and let σ and τ be permutations of A so that σ and τ are both one-to-one functions mapping A onto A. The composite function $\sigma \circ \tau$ defined schematically by

$$A \xrightarrow{\tau} A \xrightarrow{\sigma} A,$$

gives a mapping of A into A. Rather than keep the symbol $\circ$ for permutation multiplication, we will denote $\sigma \circ \tau$ by the juxtaposition $\sigma\tau$, as we have done for general groups. Now $\sigma\tau$ will be a permutation if it is one to one and onto A. *Remember that the action of $\sigma\tau$ on A must be read in right-to-left order: first apply τ and then σ.* Let us show that $\sigma\tau$ is one to one. If

$$(\sigma\tau)(a_1) = (\sigma\tau)(a_2),$$

then

$$\sigma(\tau(a_1)) = \sigma(\tau(a_2)),$$

and since σ is given to be one to one, we know that $\tau(a_1) = \tau(a_2)$. But then, since τ is one to one, this gives $a_1 = a_2$. Hence $\sigma\tau$ is one to one. To show that $\sigma\tau$ is onto A, let $a \in A$. Since σ is onto A, there exists $a' \in A$ such that $\sigma(a') = a$. Since τ is onto A, there exists $a'' \in A$ such that $\tau(a'') = a'$. Thus

$$a = \sigma(a') = \sigma(\tau(a'')) = (\sigma\tau)(a''),$$

so $\sigma\tau$ is onto A.

8.4 Example Suppose that

$$A = \{1, 2, 3, 4, 5\}$$

and that σ is the permutation given by Fig. 8.1. We write σ in a more standard notation, changing the columns to rows in parentheses and omitting the arrows, as

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix},$$

Historical Note

One of the earliest recorded studies of permutations occurs in the *Sefer Yetsirah,* or *Book of Creation,* written by an unknown Jewish author sometime before the eighth century. The author was interested in counting the various ways in which the letters of the Hebrew alphabet can be arranged. The question was in some sense a mystical one. It was believed that the letters had magical powers; therefore, suitable arrangements could subjugate the forces of nature. The actual text of the *Sefer Yetsirah* is very sparse: "Two letters build two words, three build six words, four build 24 words, five build 120, six build 720, seven build 5040." Interestingly enough, the idea of counting the arrangements of the letters of the alphabet also occurred in Islamic mathematics in the eighth and ninth centuries. By the thirteenth century, in both the Islamic and Hebrew cultures, the abstract idea of a permutation had taken root so that both Abu-l-'Abbas ibn al-Banna (1256–1321), a mathematician from Marrakech in what is now Morocco, and Levi ben Gerson, a French rabbi, philosopher, and mathematician, were able to give rigorous proofs that the number of permutations of any set of n elements is $n!$, as well as prove various results about counting combinations.

Levi and his predecessors, however, were concerned with permutations as simply arrangements of a given finite set. It was the search for solutions of polynomial equations that led Lagrange and others in the late eighteenth century to think of permutations as functions from a finite set to itself, the set being that of the roots of a given equation. And it was Augustin-Louis Cauchy (1789–1857) who developed in detail the basic theorems of permutation theory and who introduced the standard notation used in this text.

so that $\sigma(1) = 4$, $\sigma(2) = 2$, and so on. Let

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix}.$$

Then

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 3 & 2 & 4 \end{pmatrix}.$$

For example, multiplying in right-to-left order,

$$(\sigma\tau)(1) = \sigma(\tau(1)) = \sigma(3) = 5.$$ ▲

We now show that the collection of all permutations of a nonempty set A forms a group under this permutation multiplication.

8.5 Theorem Let A be a nonempty set, and let S_A be the collection of all permutations of A. Then S_A is a group under permutation multiplication.

Proof We have shown that composition of two permutations of A yields a permutation of A, so S_A is closed under permutation multiplication.

Now permutation multiplication is defined as function composition, and in Section 2, we showed that *function composition is associative.* Hence $\mathscr{G}_1$ is satisfied.

The permutation ι such that $\iota(a) = a$, for all $a \in A$ acts as identity. Therefore $\mathscr{G}_2$ is satisfied.

For a permutation σ, the inverse function, σ^{-1}, is the permutation that reverses the direction of the mapping σ, that is, $\sigma^{-1}(a)$ is the element a' of A such that $a = \sigma(a')$. The existence of exactly one such element a' is a consequence of the fact that, as a function, σ is both one to one and onto. For each $a \in A$ we have

$$\iota(a) = a = \sigma(a') = \sigma(\sigma^{-1}(a)) = (\sigma\sigma^{-1})(a)$$

and also

$$\iota(a') = a' = \sigma^{-1}(a) = \sigma^{-1}(\sigma(a')) = (\sigma^{-1}\sigma)(a'),$$

so that $\sigma^{-1}\sigma$ and $\sigma\sigma^{-1}$ are both the permutation ι. Thus $\mathcal{G}_3$ is satisfied. ◆

Warning: Some texts compute a product $\sigma\mu$ of permutations in left-to-right order, so that $(\sigma\mu)(a) = \mu(\sigma(a))$. Thus the permutation they get for $\sigma\mu$ is the one we would get by computing $\mu\sigma$. Exercise 51 asks us to check in two ways that we still get a group. If you refer to another text on this material, be sure to check its order for permutation multiplication.

There was nothing in our definition of a permutation to require that the set A be finite. However, most of our examples of permutation groups will be concerned with permutations of finite sets. Note that the *structure* of the group S_A is concerned only with the number of elements in the set A, and not what the elements in A are. If sets A and B have the same cardinality, then $S_A \simeq S_B$. To define an isomorphism $\phi : S_A \to S_B$, we let $f : A \to B$ be a one-to-one function mapping A onto B, which establishes that A and B have the same cardinality. For $\sigma \in S_A$, we let $\phi(\sigma)$ be the permutation $\bar{\sigma} \in S_B$ such that $\bar{\sigma}(f(a)) = f(\sigma(a))$ for all $a \in A$. To illustrate this for $A = \{1, 2, 3\}$ and $B = \{\#, \$, \%\}$ and the function $f : A \to B$ defined as

$$f(1) = \#, \quad f(2) = \$, \quad f(3) = \%,$$

ϕ maps

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \text{ into } \begin{pmatrix} \# & \$ & \% \\ \% & \$ & \# \end{pmatrix}.$$

We simply rename the elements of A in our two-row notation by elements in B using the renaming function f, thus renaming elements of S_A to be those of S_B. We can take $\{1, 2, 3, \cdots, \mathrm{n}\}$ to be a prototype for a finite set A of n elements.

8.6 Definition Let A be the finite set $\{1, 2, \cdots, n\}$. The group of all permutations of A is the **symmetric group on n letters**, and is denoted by S_n. ■

Note that S_n has $n!$ elements, where

$$n! = n(n-1)(n-2)\cdots(3)(2)(1).$$

Two Important Examples

8.7 Example An interesting example for us is the group S_3 of $3! = 6$ elements. Let the set A be $\{1, 2, 3\}$. We list the permutations of A and assign to each a subscripted Greek letter for a name.

The reasons for the choice of names will be clear later. Let

$$\rho_0 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \qquad \mu_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix},$$

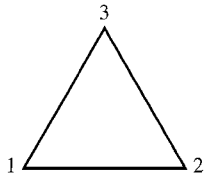
$$\rho_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \qquad \mu_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix},$$

$$\rho_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \qquad \mu_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}.$$

8.8 Table

	ρ_0	ρ_1	ρ_2	μ_1	μ_2	μ_3
ρ_0	ρ_0	ρ_1	ρ_2	μ_1	μ_2	μ_3
ρ_1	ρ_1	ρ_2	ρ_0	μ_3	μ_1	μ_2
ρ_2	ρ_2	ρ_0	ρ_1	μ_2	μ_3	μ_1
μ_1	μ_1	μ_2	μ_3	ρ_0	ρ_1	ρ_2
μ_2	μ_2	μ_3	μ_1	ρ_2	ρ_0	ρ_1
μ_3	μ_3	μ_1	μ_2	ρ_1	ρ_2	ρ_0

The multiplication table for S_3 is shown in Table 8.8. Note that this group is not abelian! We have seen that any group of at most 4 elements is abelian. Later we will see that a group of 5 elements is also abelian. Thus S_3 has minimum order for any nonabelian group. ▲



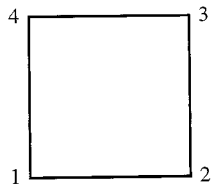
8.9 Figure

There is a natural correspondence between the elements of S_3 in Example 8.7 and the ways in which two copies of an equilateral triangle with vertices 1, 2, and 3 (see Fig. 8.9 can be placed, one covering the other with vertices on top of vertices. For this reason, S_3 is also the **group D_3 of symmetries of an equilateral triangle**. Naively, we used ρ_i for *rotations* and μ_i for *mirror images* in bisectors of angles. The notation D_3 stands for the third dihedral group. The ***n*th dihedral group** D_n is the group of symmetries of the regular n-gon. See Exercise 44.[†]

Note that we can consider the elements of S_3 to *act* on the triangle in Fig. 8.9. See the discussion at the start of this section.

8.10 Example Let us form the dihedral group D_4 of permutations corresponding to the ways that two copies of a square with vertices 1, 2, 3, and 4 can be placed, one covering the other with vertices on top of vertices (see Fig. 8.11). D_4 will then be the **group of symmetries of the square**. It is also called the **octic group**. Again, we choose seemingly arbitrary

[†] Many people denote the nth dihedral group by D_{2n} rather than by D_n since the order of the group is $2n$.



8.11 Figure

notation that we shall explain later. Naively, we are using ρ_i for *rotations*, μ_i for *mirror images* in perpendicular bisectors of sides, and δ_i for *diagonal flips*. There are eight permutations involved here. Let

$$\rho_0 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}, \qquad \mu_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix},$$

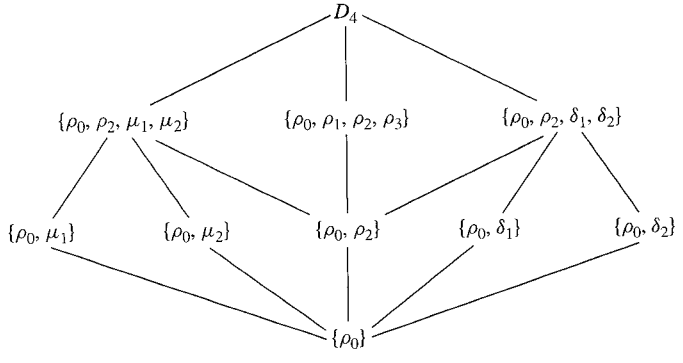
$$\rho_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}, \qquad \mu_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix},$$

$$\rho_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}, \qquad \delta_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix},$$

$$\rho_3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}, \qquad \delta_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix}.$$

8.12 Table

	ρ_0	ρ_1	ρ_2	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_0	ρ_0	ρ_1	ρ_2	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_1	ρ_1	ρ_2	ρ_3	ρ_0	δ_1	δ_2	μ_2	μ_1
ρ_2	ρ_2	ρ_3	ρ_0	ρ_1	μ_2	μ_1	δ_2	δ_1
ρ_3	ρ_3	ρ_0	ρ_1	ρ_2	δ_2	δ_1	μ_1	μ_2
μ_1	μ_1	δ_2	μ_2	δ_1	ρ_0	ρ_2	ρ_3	ρ_1
μ_2	μ_2	δ_1	μ_1	δ_2	ρ_2	ρ_0	ρ_1	ρ_3
δ_1	δ_1	μ_1	δ_2	μ_2	ρ_1	ρ_3	ρ_0	ρ_2
δ_2	δ_2	μ_2	δ_1	μ_1	ρ_3	ρ_1	ρ_2	ρ_0



8.13 Figure Subgroup diagram for D_4.

The table for D_4 is given in Table 8.12. Note that D_4 is again nonabelian. This group is simply beautiful. It will provide us with nice examples for many concepts we will introduce in group theory. Look at the lovely symmetries in that table! Finally, we give in Fig. 8.13 the subgroup diagram for the subgroups of D_4. Look at the lovely symmetries in that diagram! ▲

Cayley's Theorem

Look at any group table in the text. Note how each row of the table gives a permutation of the set of elements of the group, as listed at the top of the table. Similarly, each column of the table gives a permutation of the group set, as listed at the left of the table. In view of these observations, it is not surprising that at least every finite group G is isomorphic to a subgroup of the group S_G of all permutations of G. The same is true for infinite groups; Cayley's theorem states that *every* group is isomorphic to some group consisting of permutations under permutation multiplication. This is a nice and intriguing result, and is a classic of group theory. At first glance, the theorem might seem to be a tool to answer *all* questions about groups. What it really shows is the generality of groups of permutations. Examining subgroups of all permutation groups S_A for sets A of all sizes would be a tremendous task. Cayley's theorem does show that if a counterexample exists to some conjecture we have made about groups, then some group of permutations will provide the counterexample.

We now proceed to the proof of Cayley's theorem, starting with a definition and then a lemma that is important in its own right.

Historical Note

Arthur Cayley (1821–1895) gave an abstract-sounding definition of a group in a paper of 1854: "A set of symbols, $1, \alpha, \beta, \cdots$, all of them different and such that the product of any two of them (no matter in what order) or the product of any one of them into itself, belongs to the set, is said to be a group." He then proceeded to define a group table and note that every line and column of the table "will contain all the symbols $1, \alpha, \beta, \cdots$." Cayley's symbols, however, always represented operations on sets; it does not seem that he was aware of any other kind of group. He noted, for instance, that the four matrix operations $1, \alpha =$ inversion, $\beta =$ transposition, and $\gamma = \alpha\beta$, form, abstractly, the non-cyclic group of four elements. In any case, his definition went unnoticed for a quarter of a century.

This paper of 1854 was one of about 300 written during the 14 years Cayley was practicing law, being unable to find a suitable teaching post. In 1863, he finally became a professor at Cambridge. In 1878, he returned to the theory of groups by publishing four papers, in one of which he stated Theorem 8.16 of this text; his "proof" was simply to notice from the group table that multiplication by any group element permuted the group elements. However, he wrote, "this does not in any wise show that the best or the easiest mode of treating the general problem [of finding all groups of a given order] is thus to regard it as a problem of [permutations]. It seems clear that the better course is to consider the general problem in itself."

The papers of 1878, unlike the earlier one, found a receptive audience; in fact, they were an important influence on Walter Van Dyck's 1882 axiomatic definition of an abstract group, the definition that led to the development of abstract group theory.

8.14 Definition Let $f : A \to B$ be a function and let H be a subset of A. The **image of H under f** is $\{f(h) \mid h \in H\}$ and is denoted by $f[H]$. ■

8.15 Lemma Let G and G' be groups and let $\phi : G \to G'$ be a one-to-one function such that $\phi(xy) = \phi(x)\phi(y)$ for all $x, y \in G$. Then $\phi[G]$ is a subgroup of G' and ϕ provides an isomorphism of G with $\phi[G]$.

Proof We show the conditions for a subgroup given in Theorem 8.14 are satisfied by $\phi[G]$. Let $x', y' \in \phi[G]$. Then there exist $x, y \in G$ such that $\phi(x) = x'$ and $\phi(y) = y'$. By hypothesis, $\phi(xy) = \phi(x)\phi(y) = x'y'$, showing that $x'y' \in \phi[G]$. We have shown that $\phi[G]$ is closed under the operation of G'.

Let e' be the identity of G'. Then

$$e'\phi(e) = \phi(e) = \phi(ee) = \phi(e)\phi(e).$$

Cancellation in G' shows that $e' = \phi(e)$ so $e' \in \phi[G]$.

For $x' \in \phi[G]$ where $x' = \phi(x)$, we have

$$e' = \phi(e) = \phi(xx^{-1}) = \phi(x)\phi(x^{-1}) = x'\phi(x^{-1}),$$

which shows that $x'^{-1} = \phi(x^{-1}) \in \phi[G]$. This completes the demonstration that $\phi[G]$ is a subgroup of G'.

That ϕ provides an isomorphism of G with $\phi[G]$ now follows at once because ϕ provides a one-to-one map of G *onto* $\phi[G]$ such that $\phi(xy) = \phi(x)\phi(y)$ for all $x, y \in G$. ◆

8.16 Theorem **(Cayley's Theorem)** Every group is isomorphic to a group of permutations.

Proof Let G be a group. We show that G is isomorphic to a subgroup of S_G. By Lemma 8.15, we need only to define a one-to-one function $\phi : G \to S_G$ such that $\phi(xy) = \phi(x)\phi(y)$ for all $x, y \in G$. For $x \in G$, let $\lambda_x : G \to G$ be defined by $\lambda_x(g) = xg$ for all $g \in G$. (We think of λ_x as performing *left multiplication* by x.) The equation $\lambda_x(x^{-1}c) = x(x^{-1}c) = c$ for all $c \in G$ shows that λ_x maps G onto G. If $\lambda_x(a) = \lambda_x(b)$, then $xa = xb$ so $a = b$ by cancellation. Thus λ_x is also one to one, and is a permutation of G. We now define $\phi : G \to S_G$ by defining $\phi(x) = \lambda_x$ for all $x \in G$.

To show that ϕ is one to one, suppose that $\phi(x) = \phi(y)$. Then $\lambda_x = \lambda_y$ as functions mapping G into G. In particular $\lambda_x(e) = \lambda_y(e)$, so $xe = ye$ and $x = y$. Thus ϕ is one to one. It only remains to show that $\phi(xy) = \phi(x)\phi(y)$, that is, that $\lambda_{xy} = \lambda_x\lambda_y$. Now for any $g \in G$, we have $\lambda_{xy}(g) = (xy)g$. Permutation multiplication is function composition, so $(\lambda_x\lambda_y)(g) = \lambda_x(\lambda_y(g)) = \lambda_x(yg) = x(yg)$. Thus by associativity, $\lambda_{xy} = \lambda_x\lambda_y$. ◆

For the proof of the theorem, we could have considered equally well the permutations ρ_x of G defined by

$$\rho_x(g) = gx$$

for $g \in G$. (We can think of ρ_x as meaning *right multiplication* by x.) Exercise 52 shows that these permutations form a subgroup of S_G, again isomorphic to G, but provided by

a map $\mu : G \to S_G$ defined by

$$\mu(x) = \rho_{x^{-1}}.$$

8.17 Definition The map ϕ in the proof of Theorem 8.16 is the **left regular representation** of G, and the map μ in the preceding comment is the **right regular representation** of G. ■

8.18 Example Let us compute the left regular representation of the group given by the group table, Table 8.19. By "compute" we mean give the elements for the left regular representation and the group table. Here the elements are

$$\lambda_e = \begin{pmatrix} e & a & b \\ e & a & b \end{pmatrix}, \quad \lambda_a = \begin{pmatrix} e & a & b \\ a & b & e \end{pmatrix}, \quad \text{and} \quad \lambda_b = \begin{pmatrix} e & a & b \\ b & e & a \end{pmatrix}.$$

The table for this representation is just like the original table with x renamed λ_x, as seen in Table 8.20. For example,

$$\lambda_a\lambda_b = \begin{pmatrix} e & a & b \\ a & b & e \end{pmatrix}\begin{pmatrix} e & a & b \\ b & e & a \end{pmatrix} = \begin{pmatrix} e & a & b \\ e & a & b \end{pmatrix} = \lambda_e.$$ ▲

8.19 Table

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

8.20 Table

	λ_e	λ_a	λ_b
λ_e	λ_e	λ_a	λ_b
λ_a	λ_a	λ_b	λ_e
λ_b	λ_b	λ_e	λ_a

For a finite group given by a group table, ρ_a is the permutation of the elements corresponding to their order in the column under a at the very top, and λ_a is the permutation corresponding to the order of the elements in the row opposite a at the extreme left. The notations ρ_a and λ_a were chosen to suggest right and left multiplication by a, respectively.

EXERCISES 8

Computation

In Exercises 1 through 5, compute the indicated product involving the following permutations in S_6:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}, \quad \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 1 & 3 & 6 & 5 \end{pmatrix}, \quad \mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 2 & 4 & 3 & 1 & 6 \end{pmatrix}.$$

1. $\tau\sigma$ **2.** $\tau^2\sigma$ **3.** $\mu\sigma^2$ **4.** $\sigma^{-2}\tau$ **5.** $\sigma^{-1}\tau\sigma$

In Exercises 6 through 9, compute the expressions shown for the permutations σ, τ and μ defined prior to Exercise 1.

6. $|\langle\sigma\rangle|$ **7.** $|\langle\tau^2\rangle|$ **8.** σ^{100} **9.** μ^{100}

10. Partition the following collection of groups into subcollections of isomorphic groups. Here a * superscript means all nonzero elements of the set.

$\mathbb{Z}$ under addition	S_2
$\mathbb{Z}_6$	$\mathbb{R}^*$ under multiplication
$\mathbb{Z}_2$	$\mathbb{R}^+$ under multiplication
S_6	$\mathbb{Q}^*$ under multiplication
$17\mathbb{Z}$ under addition	$\mathbb{C}^*$ under multiplication
$\mathbb{Q}$ under addition	The subgroup $\langle \pi \rangle$ of $\mathbb{R}^*$ under multiplication
$3\mathbb{Z}$ under addition	The subgroup G of S_5 generated by $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 1 & 2 \end{pmatrix}$
$\mathbb{R}$ under addition	

Let A be a set and let $\sigma \in S_A$. For a fixed $a \in A$, the set

$$\mathcal{O}_{a,\sigma} = \{\sigma^n(a) \mid n \in \mathbb{Z}\}$$

is the **orbit** of a **under** σ. In Exercises 11 through 13, find the orbit of 1 under the permutation defined prior to Exercise 1.

11. σ **12.** τ **13.** μ

14. In Table 8.8, we used $\rho_0, \rho_1, \rho_2, \mu_1, \mu_2, \mu_3$ as the names of the 6 elements of S_3. Some authors use the notations $\epsilon, \rho, \rho^2, \phi, \rho\phi, \rho^2\phi$ for these elements, where their ϵ is our identity ρ_0, their ρ is our ρ_1, and their ϕ is our μ_1. Verify *geometrically* that their six expressions do give all of S_3.

15. With reference to Exercise 14, give a similar alternative labeling for the 8 elements of D_4 in Table 8.12.

16. Find the number of elements in the set $\{\sigma \in S_4 \mid \sigma(3) = 3\}$.

17. Find the number of elements in the set $\{\sigma \in S_5 \mid \sigma(2) = 5\}$.

18. Consider the group S_3 of Example 8.7

a. Find the cyclic subgroups $\langle \rho_1 \rangle$, $\langle \rho_2 \rangle$, and $\langle \mu_1 \rangle$ of S_3.

b. Find *all* subgroups, proper and improper, of S_3 and give the subgroup diagram for them.

19. Verify that the subgroup diagram for D_4 shown in Fig. 8.13 is correct by finding all (cyclic) subgroups generated by one element, then all subgroups generated by two elements, etc.

20. Give the multiplication table for the cyclic subgroup of S_5 generated by

$$\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 1 & 3 \end{pmatrix}.$$

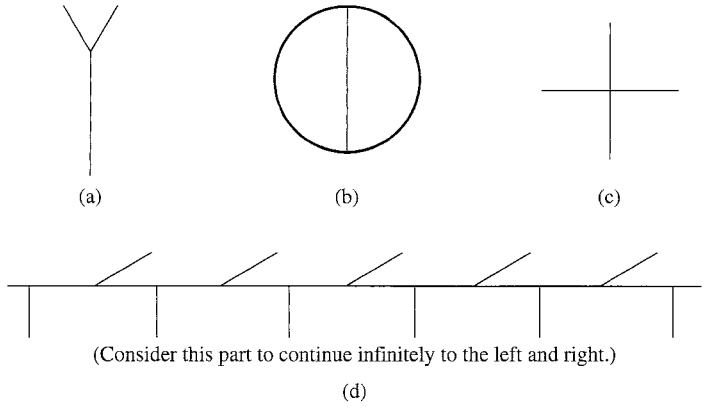
There will be six elements. Let them be $\rho, \rho^2, \rho^3, \rho^4, \rho^5$, and $\rho^0 = \rho^6$. Is this group isomorphic to S_3?

21. **a.** Verify that the six matrices

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

form a group under matrix multiplication. [*Hint*: Don't try to compute all products of these matrices. Instead, think how the column vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is transformed by multiplying it on the left by each of the matrices.]

b. What group discussed in this section is isomorphic to this group of six matrices?



8.21 Figure

22. After working Exercise 21, write down eight matrices that form a group under matrix multiplication that is isomorphic to D_4.

In this section we discussed the group of symmetries of an equilateral triangle and of a square. In Exercises 23 through 26, give a group that we have discussed in the text that is isomorphic to the group of symmetries of the indicated figure. You may want to label some special points on the figure, write some permutations corresponding to symmetries, and compute some products of permutations.

23. The figure in Fig. 8.21 (a)

24. The figure in Fig. 8.21 (b)

25. The figure in Fig. 8.21 (c)

26. The figure in Fig. 8.21 (d)

27. Compute the left regular representation of $\mathbb{Z}_4$. Compute the right regular representation of S_3 using the notation of Example 8.7.

Concepts

In Exercises 28 and 29, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

28. A *permutation* of a set S is a one-to-one map from S to S.

29. The *left regular representation* of a group G is the map of G into S_G whose value at $g \in G$ is the permutation of G that carries each $x \in G$ into gx.

In Exercises 30 through 34, determine whether the given function is a permutation of $\mathbb{R}$.

30. $f_1 : \mathbb{R} \to \mathbb{R}$ defined by $f_1(x) = x + 1$

31. $f_2 : \mathbb{R} \to \mathbb{R}$ defined by $f_2(x) = x^2$

32. $f_3 : \mathbb{R} \to \mathbb{R}$ defined by $f_3(x) = -x^3$

33. $f_4 : \mathbb{R} \to \mathbb{R}$ defined by $f_4(x) = e^x$

34. $f_5 : \mathbb{R} \to \mathbb{R}$ defined by $f_5(x) = x^3 - x^2 - 2x$

35. Mark each of the following true or false.

_____ **a.** Every permutation is a one-to-one function.

_____ **b.** Every function is a permutation if and only if it is one to one.

_____ **c.** Every function from a finite set onto itself must be one to one.

_____ **d.** Every group G is isomorphic to a subgroup of S_G.

_____ **e.** Every subgroup of an abelian group is abelian.

_____ **f.** Every element of a group generates a cyclic subgroup of the group.

_____ **g.** The symmetric group S_{10} has 10 elements.

_____ **h.** The symmetric group S_3 is cyclic.

_____ **i.** S_n is not cyclic for any n.

_____ **j.** Every group is isomorphic to some group of permutations.

36. Show by an example that every proper subgroup of a nonabelian group may be abelian.

37. Let A be a nonempty set. What type of algebraic structure mentioned previously in the text is given by the set of *all* functions mapping A into itself under function composition?

38. Indicate schematically a Cayley digraph for D_n using a generating set consisting of a rotation through $2\pi/n$ radians and a reflection (mirror image). See Exercise 44.

Proof Synopsis

39. Give a two-sentence synopsis of the proof of Cayley's theorem.

Theory

In Exercises 40 through 43, let A be a set, B a subset of A, and let b be one particular element of B. Determine whether the given set is sure to be a subgroup of S_A under the induced operation. Here $\sigma[B] = \{\sigma(x) \mid x \in B\}$.

40. $\{\sigma \in S_A \mid \sigma(b) = b\}$

41. $\{\sigma \in S_A \mid \sigma(b) \in B\}$

42. $\{\sigma \in S_A \mid \sigma[B] \subseteq B\}$

43. $\{\sigma \in S_A \mid \sigma[B] = B\}$

44. In analogy with Examples 8.7 and 8.10, consider a regular plane n-gon for $n \geq 3$. Each way that two copies of such an n-gon can be placed, with one covering the other, corresponds to a certain permutation of the vertices. The set of these permutations in a group, the **nth dihedral group** D_n, under permutation multiplication. Find the order of this group D_n. Argue *geometrically* that this group has a subgroup having just half as many elements as the whole group has.

45. Consider a cube that exactly fills a certain cubical box. As in Examples 8.7 and 8.10, the ways in which the cube can be placed into the box correspond to a certain group of permutations of the vertices of the cube. This group is the **group of rigid motions** (or **rotations) of the cube**. (It should not be confused with the *group of symmetries of the figure*, which will be discussed in the exercises of Section 12.) How many elements does this group have? Argue *geometrically* that this group has at least three different subgroups of order 4 and at least four different subgroups of order 3.

46. Show that S_n is a nonabelian group for $n \geq 3$.

47. Strengthening Exercise 46, show that if $n \geq 3$, then the only element of σ of S_n satisfying $\sigma\gamma = \gamma\sigma$ for all $\gamma \in S_n$ is $\sigma = \iota$, the identity permutation.

48. Orbits were defined before Exercise 11. Let $a, b \in A$ and $\sigma \in S_A$. Show that if $\mathcal{O}_{a,\sigma}$ and $\mathcal{O}_{b,\sigma}$ have an element in common, then $\mathcal{O}_{a,\sigma} = \mathcal{O}_{b,\sigma}$.

49. If A is a set, then a subgroup H of S_A is **transitive on** A if for each $a, b \in A$ there exists $\sigma \in H$ such that $\sigma(a) = b$. Show that if A is a nonempty finite set, then there exists a finite cyclic subgroup H of S_A with $|H| = |A|$ that is transitive on A.

50. Referring to the definition before Exercise 11 and to Exercise 49, show that for $\sigma \in S_A$, $\langle\sigma\rangle$ is transitive on A if and only if $\mathcal{O}_{a,\sigma} = A$ for some $a \in A$.

51. (See the warning on page 78). Let G be a group with binary operation $*$. Let G' be the same set as G, and define a binary operation $*'$ on G' by $x *' y = y * x$ for all $x, y \in G'$.

a. (Intuitive argument that G' under $*'$ is a group.) Suppose the front wall of your class room were made of transparent glass, and that all possible products $a * b = c$ and all possible instances $a * (b * c) =$

$(a * b) * c$ of the associative property for G under $*$ were written on the wall with a magic marker. What would a person see when looking at the other side of the wall from the next room in front of yours?

b. Show from the mathematical definition of $*'$ that G' is a group under $*'$.

52. Let G be a group. Prove that the permutations $\rho_a : G \to G$, where $\rho_a(x) = xa$ for $a \in G$ and $x \in G$, do form a group isomorphic to G.

53. A **permutation matrix** is one that can be obtained from an identity matrix by reordering its rows. If P is an $n \times n$ permutation matrix and A is any $n \times n$ matrix and $C = PA$, then C can be obtained from A by making precisely the same reordering of the rows of A as the reordering of the rows which produced P from I_n.

a. Show that every finite group of order n is isomorphic to a group consisting of $n \times n$ permutation matrices under matrix multiplication.

b. For each of the four elements e, a, b, and c in the Table 5.11 for the group V, give a specific 4×4 matrix that corresponds to it under such an isomorphism.

SECTION 9 Orbits, Cycles, and the Alternating Groups

Orbits

Each permutation σ of a set A determines a natural partition of A into cells with the property that $a, b \in A$ are in the same cell if and only if $b = \sigma^n(a)$ for some $n \in \mathbb{Z}$. We establish this partition using an appropriate equivalence relation:

$$\text{For } a, b \in A, \text{ let } a \sim b \text{ if and only if } b = \sigma^n(a) \text{ for some } n \in \mathbb{Z}. \tag{1}$$

We now check that $\sim$ defined by Condition (1) is indeed an equivalence relation.

Reflexive Clearly $a \sim a$ since $a = \iota(a) = \sigma^0(a)$.

Symmetric If $a \sim b$, then $b = \sigma^n(a)$ for some $n \in \mathbb{Z}$. But then $a = \sigma^{-n}(b)$ and $-n \in \mathbb{Z}$, so $b \sim a$.

Transitive Suppose $a \sim b$ and $b \sim c$, then $b = \sigma^n(a)$ and $c = \sigma^m(b)$ for some $n, m \in \mathbb{Z}$. Substituting, we find that $c = \sigma^m(\sigma^n(a)) = \sigma^{n+m}(a)$, so $a \sim c$.

9.1 Definition Let σ be a permutation of a set A. The equivalence classes in A determined by the equivalence relation (1) are the **orbits of** σ. ■

9.2 Example Since the identity permutation ι of A leaves each element of A fixed, the orbits of ι are the one-element subsets of A. ▲

9.3 Example Find the orbits of the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 7 & 4 & 1 & 5 & 2 \end{pmatrix}$$

in S_8.

Solution To find the orbit containing 1, we apply σ repeatedly, obtaining symbolically

$$1 \xrightarrow{\sigma} 3 \xrightarrow{\sigma} 6 \xrightarrow{\sigma} 1 \xrightarrow{\sigma} 3 \xrightarrow{\sigma} 6 \xrightarrow{\sigma} 1 \xrightarrow{\sigma} 3 \xrightarrow{\sigma} \cdots.$$

Since σ^{-1} would simply reverse the directions of the arrows in this chain, we see that the orbit containing 1 is $\{1, 3, 6\}$. We now choose an integer from 1 to 8 not in $\{1, 3, 6\}$, say 2, and similarly find that the orbit containing 2 is $\{2, 8\}$. Finally, we find that the orbit containing 4 is $\{4, 7, 5\}$. Since these three orbits include all integers from 1 to 8, we see that the complete list of orbits of σ is

$$\{1, 3, 6\}, \quad \{2, 8\}, \quad \{4, 5, 7\}. \qquad \blacktriangle$$

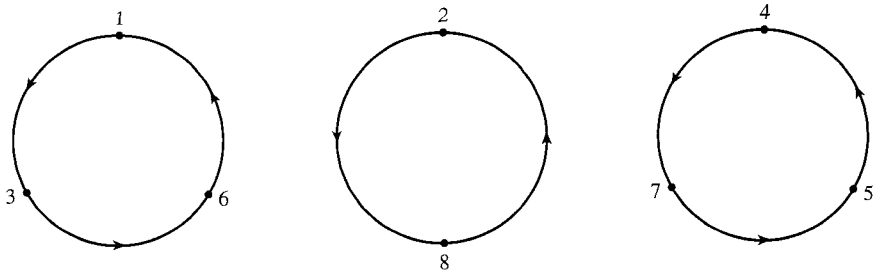
Cycles

For the remainder of this section, we consider just permutations of a finite set A of n elements. We may as well suppose that $A = \{1, 2, 3, \cdots, n\}$ and that we are dealing with elements of the symmetric group S_n.

Refer back to Example 9.3. The orbits of

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 7 & 4 & 1 & 5 & 2 \end{pmatrix} \tag{2}$$

are indicated graphically in Fig. 9.4. That is, σ acts on each integer from 1 to 8 on one of the circles by carrying it into the next integer on the circle traveled counterclockwise, in the direction of the arrows. For example, the leftmost circle indicates that $\sigma(1) = 3$, $\sigma(3) = 6$, and $\sigma(6) = 1$. Figure 9.4 is a nice way to visualize the structure of the permutation σ.

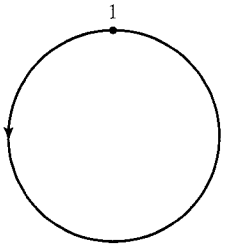


9.4 Figure

Each individual circle in Fig. 9.4 also defines, by itself, a permutation in S_8. For example, the leftmost circle corresponds to the permutation

$$\mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 2 & 6 & 4 & 5 & 1 & 7 & 8 \end{pmatrix} \tag{3}$$

that acts on 1, 3, and 6 just as σ does, but leaves the remaining integers 2, 4, 5, 7, and 8 fixed. In summary, μ has one three-element orbit $\{1, 3, 6\}$ and five one-element orbits $\{2\}, \{4\}, \{5\}, \{7\}$, and $\{8\}$. Such a permutation, described graphically by a single circle, is called a *cycle* (for circle). We consider the identity permutation to be a cycle since it can be represented by a circle having only the integer 1, as shown in Fig. 9.5. We now define the term *cycle* in a mathematically precise way.



9.5 Figure

9.6 Definition A permutation $\sigma \in S_n$ is a **cycle** if it has at most one orbit containing more than one element. The **length** of a cycle is the number of elements in its largest orbit. ■

To avoid the cumbersome notation, as in Eq. (3), for a cycle, we introduce a single-row *cyclic notation*. In cyclic notation, the cycle in Eq. (3) becomes

$$\mu = (1, 3, 6).$$

We understand by this notation that μ carries the first number 1 into the second number 3, the second number 3 into the next number 6, etc., until finally the last number 6 is carried into the first number 1. An integer not appearing in this notation for μ is understood to be left fixed by μ. Of course, the set on which μ acts, which is $\{1, 2, 3, 4, 5, 6, 7, 8\}$ in our example, must be made clear by the context.

9.7 Example Working within S_5, we see that

$$(1, 3, 5, 4) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 1 & 4 \end{pmatrix}.$$

Observe that

$$(1, 3, 5, 4) = (3, 5, 4, 1) = (5, 4, 1, 3) = (4, 1, 3, 5).$$ ▲

Of course, since cycles are special types of permutations, they can be multiplied just as any two permutations. The product of two cycles need not again be a cycle, however.

Using cyclic notation, we see that the permutation σ in Eq. (2) can be written as a product of cycles:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 7 & 4 & 1 & 5 & 2 \end{pmatrix} = (1, 3, 6)(2, 8)(4, 7, 5). \quad \textbf{(4)}$$

These cycles are **disjoint**, meaning that any integer is moved by at most one of these cycles; thus no one number appears in the notations of two different cycles. Equation (4) exhibits σ in terms of its orbits, and is a one-line description of Fig. 9.4. Every permutation in S_n can be expressed in a similar fashion as a product of the disjoint cycles corresponding to its orbits. We state this as a theorem and write out the proof.

9.8 Theorem Every permutation σ of a finite set is a product of disjoint cycles.

Proof Let $B_1, B_2, \cdots, B_r$ be the orbits of σ, and let μ_i be the cycle defined by

$$\mu_i(x) = \begin{cases} \sigma(x) & \text{for } x \in B_i \\ x & \text{otherwise.} \end{cases}$$

Clearly $\sigma = \mu_1 \mu_2 \cdots \mu_r$. Since the equivalence-class orbits $B_1, B_2, \cdots, B$, being distinct equivalence classes, are disjoint, the cycles $\mu_1, \mu_2, \cdots, \mu_r$ are disjoint also. ◆

While permutation multiplication in general is not commutative, it is readily seen that *multiplication of disjoint cycles is commutative*. Since the orbits of a permutation are unique, the representation of a permutation as a product of disjoint cycles, none of which is the identity permutation, is unique up to the order of the factors.

9.9 Example Consider the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 2 & 4 & 3 & 1 \end{pmatrix}.$$

Let us write it as a product of disjoint cycles. First, 1 is moved to 6 and then 6 to 1, giving the cycle (1, 6). Then 2 is moved to 5, which is moved to 3, which is moved to 2, or (2, 5, 3). This takes care of all elements but 4, which is left fixed. Thus

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 2 & 4 & 3 & 1 \end{pmatrix} = (1, 6)(2, 5, 3).$$

Multiplication of *disjoint* cycles is commutative, so the order of the factors (1, 6) and (2, 5, 3) is not important. ▲

You should practice multiplying permutations in cyclic notation where the cycles may or may not be disjoint. We give an example and provide further practice in Exercises 7 through 9.

9.10 Example Consider the cycles (1,4,5,6) and (2,1,5) in S_6. Multiplying, we find that

$$(1, 4, 5, 6)(2, 1, 5) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 4 & 3 & 5 & 2 & 1 \end{pmatrix}$$

and

$$(2, 1, 5)(1, 4, 5, 6) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 1 & 3 & 2 & 6 & 5 \end{pmatrix}.$$

Neither of these permutations is a cycle. ▲

Even and Odd Permutations

It seems reasonable that every reordering of the sequence $1, 2, \ldots, n$ can be achieved by repeated interchange of positions of pairs of numbers. We discuss this a bit more formally.

9.11 Definition A cycle of length 2 is a **transposition**. ■

Thus a transposition leaves all elements but two fixed, and maps each of these onto the other. A computation shows that

$$(a_1, a_2, \cdots, a_n) = (a_1, a_n)(a_1, a_{n-1}) \cdots (a_1, a_3)(a_1, a_2).$$

Therefore any cycle is a product of transpositions. We then have the following as a corollary to Theorem 9.8.

9.12 Corollary Any permutation of a finite set of at least two elements is a product of transpositions.

Naively, this corollary just states that any rearrangement of n objects can be achieved by successively interchanging pairs of them.

9.13 Example Following the remarks prior to the corollary, we see that (1, 6) (2, 5, 3) is the product (1, 6) (2, 3) (2, 5) of transpositions. ▲

9.14 Example In S_n for $n \geq 2$, the identity permutation is the product (1, 2) (1, 2) of transpositions. ▲

We have seen that every permutation of a finite set with at least two elements is a product of transpositions. The transpositions may not be disjoint, and a representation of the permutation in this way is not unique. For example, we can always insert at the beginning the transposition (1, 2) twice, because (1, 2) (1, 2) is the identity permutation. What is true is that the number of transpositions used to represent a given permutation must either always be even or always be odd. This is an important fact. We will give two proofs. The first uses a property of determinants from linear algebra. The second involves counting orbits and was suggested by David M. Bloom.

9.15 Theorem No permutation in S_n can be expressed both as a product of an even number of transpositions and as a product of an odd number of transpositions.

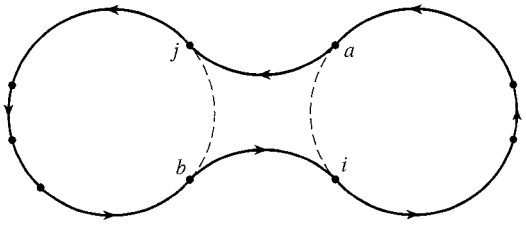
Proof 1 (From linear algebra) We remarked in Section 8 that $S_A \simeq S_B$ if A and B have the same cardinality. We work with permutations of the n rows of the $n \times n$ identity matrix I_n, rather than of the numbers $1, 2, \ldots, n$. The identity matrix has determinant 1. Interchanging any two rows of a square matrix changes the sign of the determinant. Let C be a matrix obtained by a permutation σ of the rows of I_n. If C could be obtained from I_n by both an even number and an odd number of transpositions of rows, its determinant would have to be both 1 and -1, which is impossible. Thus σ cannot be expressed both as a product of an even number and an odd number of transpositions.

Proof 2 (Counting orbits) Let $\sigma \in S_n$ and let $\tau = (i, j)$ be a transposition in S_n. We claim that the number of orbits of σ and of $\tau\sigma$ differ by 1.

Case I Suppose i and j are in different orbits of σ. Write σ as a product of disjoint cycles, the first of which contains j and the second of which contains i, symbolized by the two circles in Fig. 9.16. We may write the product of these two cycles symbolically as

$$(b, j, \times, \times, \times)(a, i, \times, \times)$$

where the symbols $\times$ denote possible other elements in these orbits.



9.16 Figure

Computing the product of the first three cycles in $\tau\sigma = (i, j)\sigma$, we obtain

$$(i, j)(b, j, \times, \times, \times)(a, i, \times, \times) = (a, j, \times, \times, \times, b, i, \times, \times).$$

The original 2 orbits have been joined to form just one in $\tau\sigma$ as symbolized in Fig. 9.16. Exercise 28 asks us to repeat the computation to show that the same thing happens if either one or both of i and j should be only element of their orbit in σ.

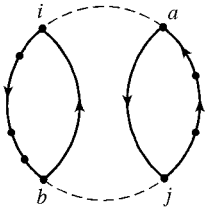
Case II Suppose i and j are in the same orbit of σ. We can then write σ as a product of disjoint cycles with the first cycle of the form

$$(a, i, \times, \times, \times, b, j, \times, \times)$$

shown symbolically by the circle in Fig. 9.17. Computing the product of the first two cycles in $\tau\sigma = (i, j)\sigma$, we obtain

$$(i, j)(a, i, \times, \times, \times, b, j, \times, \times) = (a, j, \times, \times)(b, i, \times, \times, \times).$$

The original single orbit has been split into two as symbolized in Fig. 9.17.



9.17 Figure

We have shown that the number of orbits of $\tau\sigma$ differs from the number of orbits of σ by 1. The identity permutation ι has n orbits, because each element is the only member of its orbit. Now the number of orbits of a given permutation $\sigma \in S_n$ differs from n by either an even or an odd number, but not both. Thus it is impossible to write

$$\sigma = \tau_1\tau_2\tau_3 \cdots \tau_m\iota$$

where the τ_k are transpositions in two ways, once with m even and once with m odd. ◆

9.18 Definition A permutation of a finite set is **even** or **odd** according to whether it can be expressed as a product of an even number of transpositions or the product of an odd number of transpositions, respectively. ■

9.19 Example The identity permutation ι in S_n is an even permutation since we have $\iota = (1, 2)(1, 2)$. If $n = 1$ so that we cannot form this product, we define ι to be even. On the other hand, the permutation (1, 4, 5, 6) (2, 1, 5) in S_6 can be written as

$$(1, 4, 5, 6)(2, 1, 5) = (1, 6)(1, 5)(1, 4)(2, 5)(2, 1)$$

which has five transpositions, so this is an odd permutation. ▲

The Alternating Groups

We claim that for $n \geq 2$, the number of even permutations in S_n is the same as the number of odd permutation; that is, S_n is split equally and both numbers are $(n!)/2$. To show this, let A_n be the set of even permutations in S_n and let B_n be the set of odd permutations for $n \geq 2$. We proceed to define a one-to-one function from A_n onto B_n. This is exactly what is needed to show that A_n and B_n have the same number of elements.

Let τ be any fixed transposition in S_n; it exists since $n \geq 2$. We may as well suppose that $\tau = (1, 2)$. We define a function

$$\lambda_\tau : A_n \to B_n$$

by

$$\lambda_\tau(\sigma) = \tau\sigma,$$

that is, $\sigma \in A_n$ is mapped into $(1, 2)\sigma$ by λ_τ. Observe that since σ is even, the permutation $(1, 2)\sigma$ can be expressed as a product of a (1 + even number), or odd number, of transpositions, so $(1, 2)\sigma$ is indeed in B_n. If for σ and μ in A_n it is true that $\lambda_\tau(\sigma) = \lambda_\tau(\mu)$, then

$$(1, 2)\sigma = (1, 2)\mu,$$

and since S_n is a group, we have $\sigma = \mu$. Thus λ_τ is a one-to-one function. Finally,

$$\tau = (1, 2) = \tau^{-1},$$

so if $\rho \in B_n$, then

$$\tau^{-1}\rho \in A_n,$$

and

$$\lambda_\tau(\tau^{-1}\rho) = \tau(\tau^{-1}\rho) = \rho.$$

Thus λ_τ is onto B_n. Hence the number of elements in A_n is the same as the number in B_n since there is a one-to-one correspondence between the elements of the sets.

Note that the product of two even permutations is again even. Also since $n \geq 2$, S_n has the transposition $(1, 2)$ and $\iota = (1, 2)(1, 2)$ is an even permutation. Finally, note that if σ is expressed as a product of transpositions, the product of the same transpositions taken in just the opposite order is σ^{-1}. Thus if σ is an even permutation, σ^{-1} must also be even. Referring to Theorem 5.14, we see that we have proved the following statement.

9.20 Theorem If $n \geq 2$, then the collection of all even permutations of $\{1, 2, 3, \cdots, n\}$ forms a subgroup of order $n!/2$ of the symmetric group S_n.

9.21 Definition The subgroup of S_n consisting of the even permutations of n letters is the **alternating group A_n on n letters.** ■

Both S_n and A_n are very important groups. Cayley's theorem shows that every finite group G is structurally identical to some subgroup of S_n for $n = |G|$. It can be shown that there are no formulas involving just radicals for solution of polynomial equations of degree n for $n \geq 5$. This fact is actually due to the structure of A_n, surprising as that may seem!

EXERCISES 9

Computations

In Exercises 1 through 6, find all orbits of the given permutation.

1. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 1 & 3 & 6 & 2 & 4 \end{pmatrix}$

2. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 2 & 4 & 8 & 3 & 1 & 7 \end{pmatrix}$

3. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 5 & 1 & 4 & 6 & 8 & 7 \end{pmatrix}$

4. $\sigma : \mathbb{Z} \to \mathbb{Z}$ where $\sigma(n) = n + 1$

5. $\sigma : \mathbb{Z} \to \mathbb{Z}$ where $\sigma(n) = n + 2$

6. $\sigma : \mathbb{Z} \to \mathbb{Z}$ where $\sigma(n) = n - 3$

In Exercises 7 through 9, compute the indicated product of cycles that are permutations of $\{1, 2, 3, 4, 5, 6, 7, 8\}$.

7. $(1, 4, 5)(7, 8)(2, 5, 7)$

8. $(1, 3, 2, 7)(4, 8, 6)$

9. $(1, 2)(4, 7, 8)(2, 1)(7, 2, 8, 1, 5)$

In Exercises 10 through 12, express the permutation of $\{1, 2, 3, 4, 5, 6, 7, 8\}$ as a product of disjoint cycles, and then as a product of transpositions.

10. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 2 & 6 & 3 & 7 & 4 & 5 & 1 \end{pmatrix}$

11. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 6 & 4 & 1 & 8 & 2 & 5 & 7 \end{pmatrix}$

12. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 1 & 4 & 7 & 2 & 5 & 8 & 6 \end{pmatrix}$

13. Recall that element a of a group G with identity element e has order $r > 0$ if $a^r = e$ and no smaller positive power of a is the identity. Consider the group S_8.

- **a.** What is the order of the cycle $(1, 4, 5, 7)$?
- **b.** State a theorem suggested by part (a).
- **c.** What is the order of $\sigma = (4, 5)(2, 3, 7)$? of $\tau = (1, 4)(3, 5, 7, 8)$?
- **d.** Find the order of each of the permutations given in Exercises 10 through 12 by looking at its decomposition into a product of disjoint cycles.
- **e.** State a theorem suggested by parts (c) and (d). [*Hint:* The important words you are looking for are *least common multiple*.]

In Exercises 14 through 18, find the maximum possible order for an element of S_n for the given value of n.

14. $n = 5$ **15.** $n = 6$ **16.** $n = 7$ **17.** $n = 10$ **18.** $n = 15$

19. Figure 9.22 shows a Cayley digraph for the alternating group A_4 using the generating set $S = \{(1, 2, 3), (1, 2)(3, 4)\}$. Continue labeling the other nine vertices with the elements of A_4, expressed as a product of disjoint cycles.

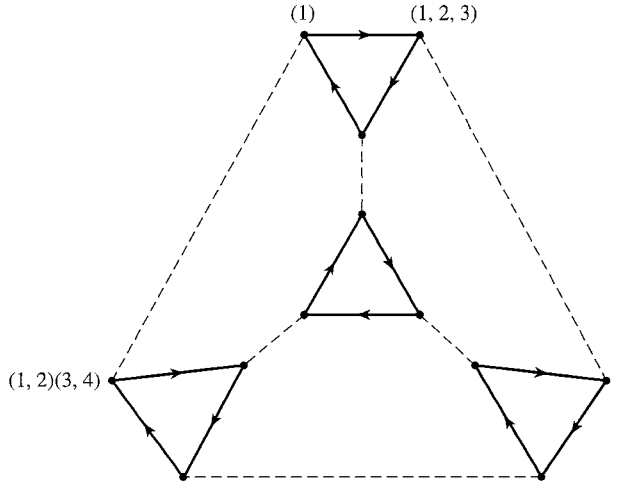
Concepts

In Exercises 20 through 22, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

20. For a permutation σ of a set A, an *orbit* of σ is a nonempty minimal subset of A that is mapped onto itself by σ.

21. A *cycle* is a permutation having only one orbit.

22. The *alternating group* is the group of all even permutations.



9.22 Figure

23. Mark each of the following true or false.

_____ **a.** Every permutation is a cycle.

_____ **b.** Every cycle is a permutation.

_____ **c.** The definition of even and odd permutations could have been given equally well before Theorem 9.15.

_____ **d.** Every nontrivial subgroup H of S_9 containing some odd permutation contains a transposition.

_____ **e.** A_5 has 120 elements.

_____ **f.** S_n is not cyclic for any $n \geq 1$.

_____ **g.** A_3 is a commutative group.

_____ **h.** S_7 is isomorphic to the subgroup of all those elements of S_8 that leave the number 8 fixed.

_____ **i.** S_7 is isomorphic to the subgroup of all those elements of S_8 that leave the number 5 fixed.

_____ **j.** The odd permutations in S_8 form a subgroup of S_8.

24. Which of the permutations in S_3 of Example 8.7 are even permutations? Give the table for the alternating group A_3.

Proof Synopsis

25. Give a one-sentence synopsis of Proof 1 of Theorem 9.15.

26. Give a two-sentence synopsis of Proof 2 of Theorem 9.15.

Theory

27. Prove the following about S_n if $n \geq 3$.

a. Every permutation in S_n can be written as a product of at most $n - 1$ transpositions.

b. Every permutation in S_n that is not a cycle can be written as a product of at most $n - 2$ transpositions.

c. Every odd permutation in S_n can be written as a product of $2n + 3$ transpositions, and every even permutation as a product of $2n + 8$ transpositions.

28. **a.** Draw a figure like Fig. 9.16 to illustrate that if i and j are in different orbits of σ and $\sigma(i) = i$, then the number of orbits of $(i, j)\sigma$ is one less than the number of orbits of σ.

b. Repeat part (a) if $\sigma(j) = j$ also.

29. Show that for every subgroup H of S_n for $n \geq 2$, either all the permutations in H are even or exactly half of them are even.

30. Let σ be a permutation of a set A. We shall say "σ **moves** $a \in A$" if $\sigma(a) \neq a$. If A is a finite set, how many elements are moved by a cycle $\sigma \in S_A$ of length n?

31. Let A be an infinite set. Let H be the set of all $\sigma \in S_A$ such that the number of elements moved by σ (see Exercise 30) is finite. Show that H is a subgroup of S_n.

32. Let A be an infinite set. Let K be the set of all $\sigma \in S_A$ that move (see Exercise 30) at most 50 elements of A. Is K a subgroup of S_A? Why?

33. Consider S_n for a fixed $n \geq 2$ and let σ be a fixed odd permutation. Show that every odd permutation in S_n is a product of σ and some permutation in A_n.

34. Show that if σ is a cycle of odd length, then σ^2 is a cycle.

35. Following the line of thought opened by Exercise 34, complete the following with a condition involving n and r so that the resulting statement is a theorem:

If σ is a cycle of length n, then σ^r is also a cycle if and only if . . .

36. Let G be a group and let a be a fixed element of G. Show that the map $\lambda_a : G \to G$, given by $\lambda_a(g) = ag$ for $g \in G$, is a permutation of the set G.

37. Referring to Exercise 36, show that $H = \{\lambda_a \mid a \in G\}$ is a subgroup of S_G, the group of all permutations of G.

38. Referring to Exercise 49 of Section 8, show that H of Exercise 37 is transitive on the set G. [*Hint:* This is an immediate corollary of one of the theorems in Section 4.]

39. Show that S_n is generated by $\{(1, 2), (1, 2, 3, \cdots, n)\}$. [*Hint:* Show that as r varies, $(1, 2, 3, \cdots, n)^r(1, 2)(1, 2, 3, \cdots, n)^{n-r}$ gives all the transpositions $(1, 2), (2, 3), (3, 4), \cdots, (n-1, n), (n, 1)$. Then show that any transposition is a product of some of these transpositions and use Corollary 9.12]

SECTION 10 Cosets and the Theorem of Lagrange

You may have noticed that the order of a subgroup H of a finite group G seems always to be a divisor of the order of G. This is the theorem of Lagrange. We shall prove it by exhibiting a partition of G into cells, all having the same size as H. Thus if there are r such cells, we will have

$$r(\text{order of } H) = (\text{order of } G)$$

from which the theorem follows immediately. The cells in the partition will be called *cosets of* H, and they are important in their own right. In Section 14, we will see that if H satisfies a certain property, then each coset can be regarded as an element of a group in a very natural way. We give some indication of such *coset groups* in this section to help you develop a feel for the topic.

Cosets

Let H be a subgroup of a group G, which may be of finite or infinite order. We exhibit two partitions of G by defining two equivalence relations, $\sim_L$ and $\sim_R$ on G.

10.1 Theorem Let H be a subgroup of G. Let the relation $\sim_L$ be defined on G by

$$a \sim_L b \qquad \text{if and only if} \qquad a^{-1}b \in H.$$

Let $\sim_R$ be defined by

$$a \sim_R b \qquad \text{if and only if} \qquad ab^{-1} \in H.$$

Then $\sim_L$ and $\sim_R$ are both equivalence relations on G.

Proof We show that $\sim_L$ is an equivalence relation, and leave the proof for $\sim_R$ to Exercise 26. When reading the proof, notice how we must constantly make use of the fact that H is a *subgroup* of G.

Reflexive Let $a \in G$. Then $a^{-1}a = e$ and $e \in H$ since H is a subgroup. Thus $a \sim_L a$.

Symmetric Suppose $a \sim_L b$. Then $a^{-1}b \in H$. Since H is a subgroup, $(a^{-1}b)^{-1}$ is in H and $(a^{-1}b)^{-1} = b^{-1}a$, so $b^{-1}a$ is in H and $b \sim_L a$.

Transitive Let $a \sim_L b$ and $b \sim_L c$. Then $a^{-1}b \in H$ and $b^{-1}c \in H$. Since H is a subgroup, $(a^{-1}b)(b^{-1}c) = a^{-1}c$ is in H, so $a \sim_L c$. ◆

The equivalence relation $\sim_L$ in Theorem 10.1 defines a partition of G, as described in Theorem 0.22. Let's see what the cells in this partition look like. Suppose $a \in G$. The cell containing a consists of all $x \in G$ such that $a \sim_L x$, which means all $x \in G$ such that $a^{-1}x \in H$. Now $a^{-1}x \in H$ if and only if $a^{-1}x = h$ for some $h \in H$, or equivalently, if and only if $x = ah$ for some $h \in H$. Therefore the cell containing a is $\{ah \mid h \in H\}$, which we denote by aH. If we go through the same reasoning for the equivalence relation $\sim_R$ defined by H, we find the cell in this partition containing $a \in G$ is $Ha = \{ha \mid h \in H\}$. Since G need not be abelian, we have no reason to expect aH and Ha to be the same subset of G. We give a formal definition.

10.2 Definition Let H be a subgroup of a group G. The subset $aH = \{ah \mid h \in H\}$ of G is the **left coset** of H containing a, while the subset $Ha = \{ha \mid h \in H\}$ is the **right coset** of H containing a. ■

10.3 Example Exhibit the left cosets and the right cosets of the subgroup $3\mathbb{Z}$ of $\mathbb{Z}$.

Solution Our notation here is additive, so the left coset of $3\mathbb{Z}$ containing m is $m + 3\mathbb{Z}$. Taking $m = 0$, we see that

$$3\mathbb{Z} = \{\cdots, -9, -6, -3, 0, 3, 6, 9, \cdots\}$$

is itself one of its left cosets, the coset containing 0. To find another left coset, we select an element of $\mathbb{Z}$ not in $3\mathbb{Z}$, say 1, and find the left coset containing it. We have

$$1 + 3\mathbb{Z} = \{\cdots, -8, -5, -2, 1, 4, 7, 10, \cdots\}.$$

These two left cosets, $3\mathbb{Z}$ and $1 + 3\mathbb{Z}$, do not yet exhaust $\mathbb{Z}$. For example, 2 is in neither of them. The left coset containing 2 is

$$2 + 3\mathbb{Z} = \{\cdots, -7, -4, -1, 2, 5, 8, 11, \cdots\}.$$

It is clear that these three left cosets we have found do exhaust $\mathbb{Z}$, so they constitute the partition of $\mathbb{Z}$ into left cosets of $3\mathbb{Z}$.

Since $\mathbb{Z}$ is abelian, the left coset $m + 3\mathbb{Z}$ and the right coset $3\mathbb{Z} + m$ are the same, so the partition of $\mathbb{Z}$ into right cosets is the same. ▲

We observe two things from Example 10.3.

> For a subgroup H of an abelian group G, the partition of G into left cosets of H and the partition into right cosets are the same.

Also, looking back at Examples 0.17 and 0.20, we see that the equivalence relation $\sim_R$ for the subgroup $n\mathbb{Z}$ of $\mathbb{Z}$ is the same as the relation of congruence modulo n. Recall that $h \equiv k \pmod{n}$ in $\mathbb{Z}$ if $h - k$ is divisible by n. This is the same as saying that $h + (-k)$ is in $n\mathbb{Z}$, which is relation $\sim_R$ of Theorem 10.1 in additive notation. Thus the partition of $\mathbb{Z}$ into cosets of $n\mathbb{Z}$ is the partition of $\mathbb{Z}$ into residue classes modulo n. For that reason, we often refer to the cells of this partition as *cosets modulo* $n\mathbb{Z}$. Note that we do not have to specify *left* or *right* cosets since they are the same for this abelian group $\mathbb{Z}$.

10.4 Example The group $\mathbb{Z}_6$ is abelian. Find the partition of $\mathbb{Z}_6$ into cosets of the subgroup $H = \{0, 3\}$.

Solution One coset is $\{0, 3\}$ itself. The coset containing 1 is $1 + \{0, 3\} = \{1, 4\}$. The coset containing 2 is $2 + \{0, 3\} = \{2, 5\}$. Since $\{0, 3\}$, $\{1, 4\}$, and $\{2, 5\}$ exhaust all of $\mathbb{Z}_6$, these are all the cosets. ▲

We point out a fascinating thing that we will develop in detail in Section 14. Referring back to Example 10.4, Table 10.5 gives the binary operation for $\mathbb{Z}_6$ but with elements listed in the order they appear in the cosets $\{0, 3\}$, $\{1, 4\}$, $(2, 5\}$. We shaded the table according to these cosets.

Suppose we denote these cosets by LT(light), MD(medium), and DK(dark) according to their shading. Table 10.5 then defines a binary operation on these shadings, as shown in Table 10.6. Note that if we replace LT by 0, MD by 1, and DK by 2 in Table 10.6, we obtain the table for $\mathbb{Z}_3$. Thus the table of shadings forms a group! We will see in

10.5 Table

$+_6$	0	3	1	4	2	5
0	0	3	1	4	2	5
3	3	0	4	1	5	2
1	1	4	2	5	3	0
4	4	1	5	2	0	3
2	2	5	3	0	4	1
5	5	2	0	3	1	4

10.6 Table

	LT	MD	DK
LT	LT	MD	DK
MD	MD	DK	LT
DK	DK	LT	MD

Section 14 that for a partition of an *abelian* group into cosets of a subgroup, reordering the group table according to the elements in the cosets always gives rise to such a *coset group*.

10.7 Example Table 10.8 again shows Table 8.8 for the symmetric group S_3 on three letters. Let H be the subgroup $\langle \mu_1 \rangle = \{\rho_0, \mu_1\}$ of S_3. Find the partitions of S_3 into left cosets of H, and the partition into right cosets of H.

Solution For the partition into left cosets, we have

$$\begin{aligned} H &= \{\rho_0, \mu_1\}, \\ \rho_1 H &= \{\rho_1\rho_0, \rho_1\mu_1\} = \{\rho_1, \mu_3\}, \\ \rho_2 H &= \{\rho_2\rho_0, \rho_2\mu_1\} = \{\rho_2, \mu_2\}. \end{aligned}$$

The partition into right cosets is

$$\begin{aligned} H &= \{\rho_0, \mu_1\}, \\ H\rho_1 &= \{\rho_0\rho_1, \mu_1\rho_1\} = \{\rho_1, \mu_2\}, \\ H\rho_2 &= \{\rho_0\rho_2, \mu_1\rho_2\} = \{\rho_2, \mu_3\}. \end{aligned}$$

The partition into left cosets of H is different from the partition into right cosets. For example, the left coset containing ρ_1 is $\{\rho_1, \mu_3\}$, while the right coset containing ρ_1 is $\{\rho_1, \mu_2\}$. This does not surprise us since the group S_3 is not abelian. ▲

Referring to Example 10.7, Table 10.9 gives permutation multiplication in S_3. The elements are listed in the order they appear in the left cosets $\{\rho_0, \mu_1\}$, $\{\rho_1, \mu_3\}$, $\{\rho_2, \mu_2\}$ found in that example. Again, we have shaded the table light, medium, and dark according to the coset to which the element belongs. Note the difference between this table and Table 10.5. This time, the body of the table does not split up into 2×2 blocks opposite and under the shaded cosets at the left and the top, as in Table 10.5 and we don't get a coset group. The product of a light element and a dark one may be either dark or medium.

Table 10.8 is shaded according to the two left cosets of the subgroup $\langle \rho_1 \rangle = \{\rho_0, \rho_1, \rho_2\}$ of S_3. These are also the two right cosets, even though S_3 is not abelian.

10.8 Table

	ρ_0	ρ_1	ρ_2	μ_1	μ_2	μ_3
ρ_0	ρ_0	ρ_1	ρ_2	μ_1	μ_2	μ_3
ρ_1	ρ_1	ρ_2	ρ_0	μ_3	μ_1	μ_2
ρ_2	ρ_2	ρ_0	ρ_1	μ_2	μ_3	μ_1
μ_1	μ_1	μ_2	μ_3	ρ_0	ρ_1	ρ_2
μ_2	μ_2	μ_3	μ_1	ρ_2	ρ_0	ρ_1
μ_3	μ_3	μ_1	μ_2	ρ_1	ρ_2	ρ_0

10.9 Table

	ρ_0	μ_1	ρ_1	μ_3	ρ_2	μ_2
ρ_0	ρ_0	μ_1	ρ_1	μ_3	ρ_2	μ_2
μ_1	μ_1	ρ_0	μ_2	ρ_2	μ_3	ρ_1
ρ_1	ρ_1	μ_3	ρ_2	μ_2	ρ_0	μ_1
μ_3	μ_3	ρ_1	μ_1	ρ_0	μ_2	ρ_2
ρ_2	ρ_2	μ_2	ρ_0	μ_1	ρ_1	μ_3
μ_2	μ_2	ρ_2	μ_3	ρ_1	μ_1	ρ_0

From Table 10.8 it is clear that we do have a coset group isomorphic to $\mathbb{Z}_2$ in this case. We will see in Section 14 that the left cosets of a subgroup H of a group G give rise to a coset group precisely when the partition of G into left cosets of H is the same as the partition into right cosets of H. In such a case, we may simply speak of the *cosets of* H, omitting the adjective left or right. We discuss coset groups in detail in Section 14, but we think it will be easier for you to understand them then if you experiment a bit with them now. Some of the exercises in this section are designed for such experimentation.

The Theorem of Lagrange

Let H be a subgroup of a group G. We claim that every left coset and every right coset of H have the same number of elements as H. We show this by exhibiting a *one-to-one* map of H *onto* a left coset gH of H for a fixed element g of G. If H is of finite order, this will show that gH has the same number of elements as H. If H is infinite, the existence of such a map is taken as the *definition* for equality of the size of H and the size of gH. (See Definition 0.13.)

Our choice for a one-to-one map $\phi : H \to gH$ is the natural one. Let $\phi(h) = gh$ for each $h \in H$. This map is onto gH by the definition of gH as $\{gh \mid h \in H\}$. To show that it is one to one, suppose that $\phi(h_1) = \phi(h_2)$ for h_1 and h_2 in H. Then $gh_1 = gh_2$ and by the cancellation law in the group G, we have $h_1 = h_2$. Thus ϕ is one to one.

Of course, a similar one-to-one map of H onto the right coset Hg can be constructed. (See Exercise 27.) We summarize as follows:

> Every coset (left or right) of a subgroup H of a group G has the same number of elements as H.

We can now prove the theorem of Lagrange.

10.10 Theorem **(Theorem of Lagrange)** Let H be a subgroup of a finite group G. Then the order of H is a divisor of the order of G.

Proof Let n be the order of G, and let H have order m. The preceding boxed statement shows that every coset of H also has m elements. Let r be the number of cells in the partition of G into left cosets of H. Then $n = rm$, so m is indeed a divisor of n. ◆

Note that this elegant and important theorem comes from the simple counting of cosets and the number of elements in each coset. *Never underestimate results that count something*! We continue to derive consequences of Theorem 10.10, which should be regarded as a counting theorem.

10.11 Corollary Every group of prime order is cyclic.

Proof Let G be of prime order p, and let a be an element of G different from the identity. Then the cyclic subgroup $\langle a \rangle$ of G generated by a has at least two elements, a and e. But by

Theorem 10.10, the order $m \geq 2$ of $\langle a \rangle$ must divide the prime p. Thus we must have $m = p$ and $\langle a \rangle = G$, so G is cyclic. ◆

Since every cyclic group of order p is isomorphic to $\mathbb{Z}_p$, we see that *there is only one group structure, up to isomorphism, of a given prime order* p. Now doesn't this elegant result follow easily from the theorem of Lagrange, a *counting* theorem? *Never underestimate a theorem that counts something*. Proving the preceding corollary is a favorite examination question.

10.12 Theorem The order of an element of a finite group divides the order of the group.

Proof Remembering that the order of an element is the same as the order of the cyclic subgroup generated by the element, we see that this theorem follows directly from Theorem 10.10. ◆

10.13 Definition Let H be a subgroup of a group G. The number of left cosets of H in G is the **index** $(G : H)$ **of** H **in** G. ■

The index $(G : H)$ just defined may be finite or infinite. If G is finite, then obviously $(G : H)$ is finite and $(G : H) = |G|/|H|$, since every coset of H contains $|H|$ elements. Exercise 35 shows the index $(G : H)$ could be equally well defined as the number of right cosets of H in G. We state a basic theorem concerning indices of subgroups, and leave the proof to the exercises (see Exercise 38).

10.14 Theorem Suppose H and K are subgroups of a group G such that $K \leq H \leq G$, and suppose $(H : K)$ and $(G : H)$ are both finite. Then $(G : K)$ is finite, and $(G : K) = (G : H)(H : K)$.

Theorem 10.10 shows that if there is a subgroup H of a finite group G, then the order of H divides the order of G. Is the converse true? That is, if G is a group of order n, and m divides n, is there always a subgroup of order m? We will see in the next section that this is true for abelian groups. However, A_4 can be shown to have no subgroup of order 6, which gives a counterexample for nonabelian groups.

EXERCISES 10

Computations

1. Find all cosets of the subgroup $4\mathbb{Z}$ of $\mathbb{Z}$.
2. Find all cosets of the subgroup $4\mathbb{Z}$ of $2\mathbb{Z}$.
3. Find all cosets of the subgroup $\langle 2 \rangle$ of $\mathbb{Z}_{12}$.
4. Find all cosets of the subgroup $\langle 4 \rangle$ of $\mathbb{Z}_{12}$.
5. Find all cosets of the subgroup $\langle 18 \rangle$ of $\mathbb{Z}_{36}$.
6. Find all left cosets of the subgroup $\{\rho_0, \mu_2\}$ of the group D_4 given by Table 8.12.
7. Repeat the preceding exercise, but find the right cosets this time. Are they the same as the left cosets?

8. Rewrite Table 8.12 in the order exhibited by the left cosets in Exercise 6. Do you seem to get a coset group of order 4? If so, is it isomorphic to $\mathbb{Z}_4$ or to the Klein 4-group V?

9. Repeat Exercise 6 for the subgroup $\{\rho_0, \rho_2\}$ of D_4.

10. Repeat the preceding exercise, but find the right cosets this time. Are they the same as the left coset?

11. Rewrite Table 8.12 in the order exhibited by the left cosets in Exercise 9. Do you seem to get a coset group of order 4? If so, is it isomorphic to $\mathbb{Z}_4$ or to the Klein 4-group V?

12. Find the index of $\langle 3 \rangle$ in the group $\mathbb{Z}_{24}$.

13. Find the index of $\langle \mu_1 \rangle$ in the group S_3, using the notation of Example 10.7

14. Find the index of $\langle \mu_2 \rangle$ in the group D_4 given in Table 8.12

15. Let $\sigma = (1, 2, 5, 4)(2, 3)$ in S_5. Find the index of $\langle \sigma \rangle$ in S_5.

16. Let $\mu = (1, 2, 4, 5)(3, 6)$ in S_6. Find the index of $\langle \mu \rangle$ in S_6.

Concepts

In Exercises 17 and 18, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

17. Let G be a group and let $H \subseteq G$. The *left coset of H containing a* is $aH = \{ah \mid h \in H\}$.

18. Let G be a group and let $H \leq G$. The *index of H in G* is the number of right cosets of H in G.

19. Mark each of the following true or false.

_____ **a.** Every subgroup of every group has left cosets.
_____ **b.** The number of left cosets of a subgroup of a finite group divides the order of the group.
_____ **c.** Every group of prime order is abelian.
_____ **d.** One cannot have left cosets of a finite subgroup of an infinite group.
_____ **e.** A subgroup of a group is a left coset of itself.
_____ **f.** Only subgroups of finite groups can have left cosets.
_____ **g.** A_n is of index 2 in S_n for $n > 1$.
_____ **h.** The theorem of Lagrange is a nice result.
_____ **i.** Every finite group contains an element of every order that divides the order of the group.
_____ **j.** Every finite cyclic group contains an element of every order that divides the order of the group.

In Exercises 20 through 24, give an example of the desired subgroup and group if possible. If impossible, say why it is impossible.

20. A subgroup of an abelian group G whose left cosets and right cosets give different partitions of G

21. A subgroup of a group G whose left cosets give a partition of G into just one cell

22. A subgroup of a group of order 6 whose left cosets give a partition of the group into 6 cells

23. A subgroup of a group of order 6 whose left cosets give a partition of the group into 12 cells

24. A subgroup of a group of order 6 whose left cosets give a partition of the group into 4 cells

Proof Synopsis

25. Give a one-sentence synopsis of the proof of Theorem 10.10.

Theory

26. Prove that the relation $\sim_R$ of Theorem 10.1 is an equivalence relation.

27. Let H be a subgroup of a group G and let $g \in G$. Define a one-to-one map of H onto Hg. Prove that your map is one to one and is onto Hg.

28. Let H be a subgroup of a group G such that $g^{-1}hg \in H$ for all $g \in G$ and all $h \in H$. Show that every left coset gH is the same as the right coset Hg.

29. Let H be a subgroup of a group G. Prove that if the partition of G into left cosets of H is the same as the partition into right cosets of H, then $g^{-1}hg \in H$ for all $g \in G$ and all $h \in H$. (Note that this is the converse of Exercise 28.)

Let H be a subgroup of a group G and let $a, b \in G$. In Exercises 30 through 33 prove the statement or give a counterexample.

30. If $aH = bH$, then $Ha = Hb$.

31. If $Ha = Hb$, then $b \in Ha$.

32. If $aH = bH$, then $Ha^{-1} = Hb^{-1}$.

33. If $aH = bH$, then $a^2H = b^2H$.

34. Let G be a group of order pq, where p and q are prime numbers. Show that every proper subgroup of G is cyclic.

35. Show that there are the same number of left as right cosets of a subgroup H of a group G; that is, exhibit a one-to-one map of the collection of left cosets onto the collection of right cosets. (Note that this result is obvious by counting for finite groups. Your proof must hold for any group.)

36. Exercise 29 of Section 4 showed that every finite group of even order $2n$ contains an element of order 2. Using the theorem of Lagrange, show that if n is odd, then an abelian group of order $2n$ contains precisely one element of order 2.

37. Show that a group with at least two elements but with no proper nontrivial subgroups must be finite and of prime order.

38. Prove Theorem 10.14 [*Hint:* Let $\{a_iH \mid i = 1, \cdots, r\}$ be the collection of distinct left cosets of H in G and $\{b_jK \mid j = 1, \cdots, s\}$ be the collection of distinct left cosets of K in H. Show that

$$\{(a_ib_j)K \mid i = 1, \cdots, r; j = 1, \cdots, s\}$$

is the collection of distinct left cosets of K in G.]

39. Show that if H is a subgroup of index 2 in a finite group G, then every left coset of H is also a right coset of H.

40. Show that if a group G with identity e has finite order n, then $a^n = e$ for all $a \in G$.

41. Show that every left coset of the subgroup $\mathbb{Z}$ of the additive group of real numbers contains exactly one element x such that $0 \leq x < 1$.

42. Show that the function *sine* assigns the same value to each element of any fixed left coset of the subgroup $\langle 2\pi \rangle$ of the additive group $\mathbb{R}$ of real numbers. (Thus *sine* induces a well-defined function on the set of cosets; the value of the function on a coset is obtained when we choose an element x of the coset and compute $\sin x$.)

43. Let H and K be subgroups of a group G. Define $\sim$ on G by $a \sim b$ if and only if $a = hbk$ for some $h \in H$ and some $k \in K$.

a. Prove that $\sim$ is an equivalence relation on G.

b. Describe the elements in the equivalence class containing $a \in G$. (These equivalence classes are called **double cosets.**)

44. Let S_A be the group of all permutations of the set A, and let c be one particular element of A.

a. Show that $\{\sigma \in S_A \mid \sigma(c) = c\}$ is a subgroup $S_{c,c}$ of S_A.

b. Let $d \neq c$ be another particular element of A. Is $S_{c,d} = \{\sigma \in S_A \mid \sigma(c) = d\}$ a subgroup of S_A? Why or why not?

c. Characterize the set $S_{c,d}$ of part (b) in terms of the subgroup $S_{c,c}$ of part (a).

45. Show that a finite cyclic group of order n has exactly one subgroup of each order d dividing n, and that these are all the subgroups it has.

46. The **Euler phi-function** is defined for positive integers n by $\varphi(n) = s$, where s is the number of positive integers less than or equal to n that are relatively prime to n. Use Exercise 45 to show that

$$n = \sum_{d \mid n} \varphi(d),$$

the sum being taken over all positive integers d dividing n. [*Hint:* Note that the number of generators of $\mathbb{Z}_d$ is $\varphi(d)$ by Corollary 6.16.]

47. Let G be a finite group. Show that if for each positive integer m the number of solutions x of the equation $x^m = e$ in G is at most m, then G is cyclic. [*Hint:* Use Theorem 10.12 and Exercise 46 to show that G must contain an element of order $n = |G|$.]

SECTION 11 DIRECT PRODUCTS AND FINITELY GENERATED ABELIAN GROUPS

Direct Products

Let us take a moment to review our present stockpile of groups. Starting with finite groups, we have the cyclic group $\mathbb{Z}_n$, the symmetric group S_n, and the alternating group A_n for each positive integer n. We also have the dihedral groups D_n of Section 8, and the Klein 4-group V. Of course we know that subgroups of these groups exist. Turning to infinite groups, we have groups consisting of sets of numbers under the usual addition or multiplication, as, for example, $\mathbb{Z}$, $\mathbb{R}$, and $\mathbb{C}$ under addition, and their nonzero elements under multiplication. We have the group U of complex numbers of magnitude 1 under multiplication, which is isomorphic to each of the groups $\mathbb{R}_c$ under addition modulo c, where $c \in \mathbb{R}^+$. We also have the group S_A of all permutations of an infinite set A, as well as various groups formed from matrices.

One purpose of this section is to show a way to use known groups as building blocks to form more groups. The Klein 4-group will be recovered in this way from the cyclic groups. Employing this procedure with the cyclic groups gives us a large class of abelian groups that can be shown to include all possible structure types for a finite abelian group. We start by generalizing Definition 0.4.

11.1 Definition The **Cartesian product of sets** $S_1, S_2, \cdots, S_n$ is the set of all ordered n-tuples $(a_1, a_2, \cdots, a_n)$, where $a_i \in S_i$ for $i = 1, 2, \cdots, n$. The Cartesian product is denoted by either

$$S_1 \times S_2 \times \cdots \times S_n$$

or by

$$\prod_{i=1}^{n} S_i. \qquad \blacksquare$$

We could also define the Cartesian product of an infinite number of sets, but the definition is considerably more sophisticated and we shall not need it.

Now let $G_1, G_2, \cdots, G_n$ be groups, and let us use multiplicative notation for all the group operations. Regarding the G_i as sets, we can form $\prod_{i=1}^{n} G_i$. Let us show that we can make $\prod_{i=1}^{n} G_i$ into a group by means of a binary operation of *multiplication by*

components. Note again that we are being sloppy when we use the same notation for a group as for the set of elements of the group.

11.2 Theorem Let $G_1, G_2, \cdots, G_n$ be groups. For $(a_1, a_2, \cdots, a_n)$ and $(b_1, b_2, \cdots, b_n)$ in $\prod_{i=1}^{n} G_i$, define $(a_1, a_2, \cdots, a_n)(b_1, b_2, \cdots, b_n)$ to be the element $(a_1b_1, a_2b_2, \cdots, a_nb_n)$. Then $\prod_{i=1}^{n} G_i$ is a group, the **direct product of the groups** G_i, under this binary operation.

Proof Note that since $a_i \in G_i$, $b_i \in G_i$, and G_i is a group, we have $a_ib_i \in G_i$. Thus the definition of the binary operation on $\prod_{i=1}^{n} G_i$ given in the statement of the theorem makes sense; that is, $\prod_{i=1}^{n} G_i$ is closed under the binary operation.

The associative law in $\prod_{i=1}^{n} G_i$ is thrown back onto the associative law in each component as follows:

$$\begin{aligned}
&(a_1, a_2, \cdots, a_n)[(b_1, b_2, \cdots, b_n)(c_1, c_2, \cdots, c_n)] \\
&\quad = (a_1, a_2, \cdots, a_n)(b_1c_1, b_2c_2, \cdots, b_nc_n) \\
&\quad = (a_1(b_1c_1), a_2(b_2c_2), \cdots, a_n(b_nc_n)) \\
&\quad = ((a_1b_1)c_1, (a_2b_2)c_2, \cdots, (a_nb_n)c_n) \\
&\quad = (a_1b_1, a_2b_2, \cdots, a_nb_n)(c_1, c_2, \cdots, c_n) \\
&\quad = [(a_1, a_2, \cdots, a_n)(b_1, b_2, \cdots, b_n)](c_1, c_2, \cdots, c_n).
\end{aligned}$$

If e_i is the identity element in G_i, then clearly, with multiplication by components, $(e_1, e_2, \cdots, e_n)$ is an identity in $\prod_{i=1}^{n} G_i$. Finally, an inverse of $(a_1, a_2, \cdots, a_n)$ is $(a_1^{-1}, a_2^{-1}, \cdots, a_n^{-1})$; compute the product by components. Hence $\prod_{i=1}^{n} G_i$ is a group. ◆

In the event that the operation of each G_i is commutative, we sometimes use additive notation in $\prod_{i=1}^{n} G_i$ and refer to $\prod_{i=1}^{n} G_i$ as the **direct sum of the groups** G_i. The notation $\oplus_{i=1}^{n} G_i$ is sometimes used in this case in place of $\prod_{i=1}^{n} G_i$, especially with abelian groups with operation $+$. The direct sum of abelian groups $G_1, G_2, \cdots, G_n$ may be written $G_1 \oplus G_2 \oplus \cdots \oplus G_n$. We leave to Exercise 46 the proof that a direct product of abelian groups is again abelian.

It is quickly seen that if the S_i has r_i elements for $i = 1, \cdots, n$, then $\prod_{i=1}^{n} S_i$ has $r_1r_2 \cdots r_n$ elements, for in an n-tuple, there are r_1 choices for the first component from S_1, and for each of these there are r_2 choices for the next component from S_2, and so on.

11.3 Example Consider the group $\mathbb{Z}_2 \times \mathbb{Z}_3$, which has $2 \cdot 3 = 6$ elements, namely $(0, 0)$, $(0, 1)$, $(0, 2)$, $(1, 0)$, $(1, 1)$, and $(1, 2)$. We claim that $\mathbb{Z}_2 \times \mathbb{Z}_3$ is cyclic. It is only necessary to find a generator. Let us try $(1, 1)$. Here the operations in $\mathbb{Z}_2$ and $\mathbb{Z}_3$ are written additively, so we do the same in the direct product $\mathbb{Z}_2 \times \mathbb{Z}_3$.

$$\begin{aligned}
(1, 1) &= (1, 1) \\
2(1, 1) &= (1, 1) + (1, 1) = (0, 2) \\
3(1, 1) &= (1, 1) + (1, 1) + (1, 1) = (1, 0) \\
4(1, 1) &= 3(1, 1) + (1, 1) = (1, 0) + (1, 1) = (0, 1) \\
5(1, 1) &= 4(1, 1) + (1, 1) = (0, 1) + (1, 1) = (1, 2) \\
6(1, 1) &= 5(1, 1) + (1, 1) = (1, 2) + (1, 1) = (0, 0)
\end{aligned}$$

Thus (1, 1) generates all of $\mathbb{Z}_2 \times \mathbb{Z}_3$. Since there is, up to isomorphism, only one cyclic group structure of a given order, we see that $\mathbb{Z}_2 \times \mathbb{Z}_3$ is isomorphic to $\mathbb{Z}_6$. ▲

11.4 Example Consider $\mathbb{Z}_3 \times \mathbb{Z}_3$. This is a group of nine elements. We claim that $\mathbb{Z}_3 \times \mathbb{Z}_3$ is *not* cyclic. Since the addition is by components, and since in $\mathbb{Z}_3$ every element added to itself three times gives the identity, the same is true in $\mathbb{Z}_3 \times \mathbb{Z}_3$. Thus no element can generate the group, for a generator added to itself successively could only give the identity after nine summands. We have found another group structure of order 9. A similar argument shows that $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not cyclic. Thus $\mathbb{Z}_2 \times \mathbb{Z}_2$ must be isomorphic to the Klein 4-group. ▲

The preceding examples illustrate the following theorem:

11.5 Theorem The group $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic and is isomorphic to $\mathbb{Z}_{mn}$ if and only if m and n are relatively prime, that is, the gcd of m and n is 1.

Proof Consider the cyclic subgroup of $\mathbb{Z}_m \times \mathbb{Z}_n$ generated by (1, 1) as described by Theorem 5.17. As our previous work has shown, the order of this cyclic subgroup is the smallest power of (1, 1) that gives the identity (0, 0). Here taking a power of (1, 1) in our additive notation will involve adding (1, 1) to itself repeatedly. Under addition by components, the first component $1 \in \mathbb{Z}_m$ yields 0 only after m summands, $2m$ summands, and so on, and the second component $1 \in \mathbb{Z}_n$ yields 0 only after n summands, $2n$ summands, and so on. For them to yield 0 simultaneously, the number of summands must be a multiple of both m and n. The smallest number that is a multiple of both m and n will be mn if and only if the gcd of m and n is 1; in this case, (1, 1) generates a cyclic subgroup of order mn, which is the order of the whole group. This shows that $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic of order mn, and hence isomorphic to $\mathbb{Z}_{mn}$ if m and n are relatively prime.

For the converse, suppose that the gcd of m and n is $d > 1$. The mn/d is divisible by both m and n. Consequently, for any (r, s) in $\mathbb{Z}_m \times \mathbb{Z}_n$, we have

$$\underbrace{(r, s) + (r, s) + \cdots + (r, s)}_{mn/d \text{ summands}} = (0, 0).$$

Hence no element (r, s) in $\mathbb{Z}_m \times \mathbb{Z}_n$ can generate the entire group, so $\mathbb{Z}_m \times \mathbb{Z}_n$ is not cyclic and therefore not isomorphic to $\mathbb{Z}_{mn}$. ◆

This theorem can be extended to a product of more than two factors by similar arguments. We state this as a corollary without going through the details of the proof.

11.6 Corollary The group $\prod_{i=1}^{n} \mathbb{Z}_{m_i}$ is cyclic and isomorphic to $\mathbb{Z}_{m_1 m_2 \cdots m_n}$ if and only if the numbers m_i for $i = 1, \cdots, n$ are such that the gcd of any two of them is 1.

11.7 Example The preceding corollary shows that if n is written as a product of powers of distinct prime numbers, as in

$$n = (p_1)^{n_1}(p_2)^{n_2} \cdots (p_r)^{n_r},$$

then $\mathbb{Z}_n$ is isomorphic to

$$\mathbb{Z}_{(p_1)^{n_1}} \times \mathbb{Z}_{(p_2)^{n_2}} \times \cdots \times \mathbb{Z}_{(p_r)^{n_r}}.$$

In particular, $\mathbb{Z}_{72}$ is isomorphic to $\mathbb{Z}_8 \times \mathbb{Z}_9$. ▲

We remark that changing the order of the factors in a direct product yields a group isomorphic to the original one. The names of elements have simply been changed via a permutation of the components in the n-tuples.

Exercise 47 of Section 6 asked you to define the least common multiple of two positive integers r and s as a generator of a certain cyclic group. It is straightforward to prove that the subset of $\mathbb{Z}$ consisting of all integers that are multiples of both r and s is a subgroup of $\mathbb{Z}$, and hence is a cyclic group. Likewise, the set of all common multiples of n positive integers $r_1, r_2, \cdots, r_n$ is a subgroup of $\mathbb{Z}$, and hence is cyclic.

11.8 Definition Let $r_1, r_2, \cdots, r_n$ be positive integers. Their **least common multiple** (abbreviated lcm) is the positive generator of the cyclic group of all common multiples of the r_i, that is, the cyclic group of all integers divisible by each r_i for $i = 1, 2, \cdots, n$. ■

From Definition 11.8 and our work on cyclic groups, we see that the lcm of $r_1, r_2, \cdots, r_n$ is the smallest positive integer that is a multiple of each r_i for $i = 1, 2, \cdots, n$, hence the name *least common multiple*.

11.9 Theorem Let $(a_1, a_2, \cdots, a_n) \in \prod_{i=1}^{n} G_i$. If a_i is of finite order r_i in G_i, then the order of $(a_1, a_2, \cdots, a_n)$ in $\prod_{i=1}^{n} G_i$ is equal to the least common multiple of all the r_i.

Proof This follows by a repetition of the argument used in the proof of Theorem 11.5. For a power of $(a_1, a_2, \cdots, a_n)$ to give $(e_1, e_2, \cdots, e_n)$, the power must simultaneously be a multiple of r_1 so that this power of the first component a_1 will yield e_1, a multiple of r_2, so that this power of the second component a_2 will yield e_2, and so on. ◆

11.10 Example Find the order of (8, 4, 10) in the group $\mathbb{Z}_{12} \times \mathbb{Z}_{60} \times \mathbb{Z}_{24}$.

Solution Since the gcd of 8 and 12 is 4, we see that 8 is of order $\frac{12}{4} = 3$ in $\mathbb{Z}_{12}$. (See Theorem 6.14.) Similarly, we find that 4 is of order 15 in $\mathbb{Z}_{60}$ and 10 is of order 12 in $\mathbb{Z}_{24}$. The lcm of 3, 15, and 12 is $3 \cdot 5 \cdot 4 = 60$, so (8, 4, 10) is of order 60 in the group $\mathbb{Z}_{12} \times \mathbb{Z}_{60} \times \mathbb{Z}_{24}$. ▲

11.11 Example The group $\mathbb{Z} \times \mathbb{Z}_2$ is generated by the elements (1, 0) and (0, 1). More generally, the direct product of n cyclic groups, each of which is either $\mathbb{Z}$ or $\mathbb{Z}_m$ for some positive integer m, is generated by the n n-tuples

$$(1, 0, 0, \cdots, 0), \quad (0, 1, 0, \cdots, 0), \quad (0, 0, 1, \cdots, 0), \quad \cdots, \quad (0, 0, 0, \cdots, 1).$$

Such a direct product might also be generated by fewer elements. For example, $\mathbb{Z}_3 \times \mathbb{Z}_4 \times \mathbb{Z}_{35}$ is generated by the single element (1, 1, 1). ▲

Note that if $\prod_{i=1}^{n} G_i$ is the direct product of groups G_i, then the subset

$$\bar{G}_i = \{(e_1, e_2, \cdots, e_{i-1}, a_i, e_{i+1}, \cdots, e_n) \mid a_i \in G_i\},$$

that is, the set of all n-tuples with the identity elements in all places but the ith, is a subgroup of $\prod_{i=1}^{n} G_i$. It is also clear that this subgroup $\bar{G}_i$ is naturally isomorphic to G_i; just rename

$$(e_1, e_2, \cdots, e_{i-1}, a_i, e_{i+1}, \cdots, e_n) \text{ by } a_i.$$

The group G_i is mirrored in the ith component of the elements of $\overline{G}_i$, and the e_j in the other components just ride along. We consider $\prod_{i=1}^{n} G_i$ to be the *internal direct product* of these subgroups $\overline{G}_i$. The direct product given by Theorem 11.2 is called the *external direct product* of the groups G_i. The terms *internal* and *external,* as applied to a direct product of groups, just reflect whether or not (respectively) we are regarding the component groups as subgroups of the product group. We shall usually omit the words *external* and *internal* and just say *direct product.* Which term we mean will be clear from the context.

Historical Note

In his *Disquisitiones Arithmeticae,* Carl Gauss demonstrated various results in what is today the theory of abelian groups in the context of number theory. Not only did he deal extensively with equivalence classes of quadratic forms, but he also considered residue classes modulo a given integer. Although he noted that results in these two areas were similar, he did not attempt to develop an abstract theory of abelian groups.

In the 1840s, Ernst Kummer in dealing with ideal complex numbers noted that his results were in many respects analogous to those of Gauss. (See the Historical Note in Section 26.) But it was Kummer's student Leopold Kronecker (see the Historical Note in Section 29) who finally realized that an abstract theory could be developed out of the analogies. As he wrote in 1870, "these principles [from the work of Gauss and Kummer] belong to a more general, abstract realm of ideas. It is therefore appropriate to free their development from all unimportant restrictions, so that one can spare oneself from the necessity of repeating the same argument in different cases. This advantage already appears in the development itself, and the presentation gains in simplicity, if it is given in the most general admissible manner, since the most important features stand out with clarity." Kronecker then proceeded to develop the basic principles of the theory of finite abelian groups and was able to state and prove a version of Theorem 11.12 restricted to finite groups.

The Structure of Finitely Generated Abelian Groups

Some theorems of abstract algebra are easy to understand and use, although their proofs may be quite technical and time-consuming to present. This is one section in the text where we explain the meaning and significance of a theorem but omit its proof. The meaning of any theorem whose proof we omit is well within our understanding, and we feel we should be acquainted with it. It would be impossible for us to meet some of these fascinating facts in a one-semester course if we were to insist on wading through complete proofs of all theorems. The theorem that we now state gives us complete structural information about all sufficiently small abelian groups, in particular, about all finite abelian groups.

11.12 Theorem **(Fundamental Theorem of Finitely Generated Abelian Groups)** Every finitely generated abelian group G is isomorphic to a direct product of cyclic groups in the form

$$\mathbb{Z}_{(p_1)^{r_1}} \times \mathbb{Z}_{(p_2)^{r_2}} \times \cdots \times \mathbb{Z}_{(p_n)^{r_n}} \times \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z},$$

where the p_i are primes, not necessarily distinct, and the r_i are positive integers. The direct product is unique except for possible rearrangement of the factors; that is, the number (**Betti number** of G) of factors $\mathbb{Z}$ is unique and the prime powers $(p_i)^{r_i}$ are unique.

Proof The proof is omitted here. ◆

11.13 Example Find all abelian groups, up to isomorphism, of order 360. The phrase *up to isomorphism* signifies that any abelian group of order 360 should be structurally identical (isomorphic) to one of the groups of order 360 exhibited.

Solution We make use of Theorem 11.12. Since our groups are to be of the finite order 360, no factors $\mathbb{Z}$ will appear in the direct product shown in the statement of the theorem.

First we express 360 as a product of prime powers $2^3 3^2 5$. Then using Theorem 11.12, we get as possibilities

1. $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$
2. $\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$
3. $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$
4. $\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_5$
5. $\mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$
6. $\mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_5$

Thus there are six different abelian groups (up to isomorphism) of order 360. ▲

Applications

We conclude this section with a sampling of the many theorems we could now prove regarding abelian groups.

11.14 Definition A group G is **decomposable** if it is isomorphic to a direct product of two proper nontrivial subgroups. Otherwise G is **indecomposable**. ■

11.15 Theorem The finite indecomposable abelian groups are exactly the cyclic groups with order a power of a prime.

Proof Let G be a finite indecomposable abelian group. Then by Theorem 11.12, G is isomorphic to a direct product of cyclic groups of prime power order. Since G is indecomposable, this direct product must consist of just one cyclic group whose order is a power of a prime number.

Conversely, let p be a prime. Then $\mathbb{Z}_{p^r}$ is indecomposable, for if $\mathbb{Z}_{p^r}$ were isomorphic to $\mathbb{Z}_{p^i} \times \mathbb{Z}_{p^j}$, where $i + j = r$, then every element would have an order at most $p^{\max(i,j)} < p^r$. ◆

11.16 Theorem If m divides the order of a finite abelian group G, then G has a subgroup of order m.

Proof By Theorem 11.12, we can think of G as being

$$\mathbb{Z}_{(p_1)^{r_1}} \times \mathbb{Z}_{(p_2)^{r_2}} \times \cdots \times \mathbb{Z}_{(p_n)^{r_n}},$$

where not all primes p_i need be distinct. Since $(p_1)^{r_1}(p_2)^{r_2}\cdots(p_n)^{r_n}$ is the order of G, then m must be of the form $(p_1)^{s_1}(p_2)^{s_2}\cdots(p_n)^{s_n}$, where $0 \le s_i \le r_i$. By Theorem 6.14, $(p_i)^{r_i-s_i}$ generates a cyclic subgroup of $\mathbb{Z}_{(p_i)^{r_i}}$ of order equal to the quotient of $(p_i)^{r_i}$ by the gcd of $(p_i)^{r_i}$ and $(p_i)^{r_i-s_i}$. But the gcd of $(p_i)^{r_i}$ and $(p_i)^{r_i-s_i}$ is $(p_i)^{r_i-s_i}$. Thus $(p_i)^{r_i-s_i}$ generates a cyclic subgroup $\mathbb{Z}_{(p_i)^{r_i}}$ of order

$$[(p_i)^{r_i}]/[(p_i)^{r_i-s_i}] = (p_i)^{s_i}.$$

Recalling that $\langle a\rangle$ denotes the cyclic subgroup generated by a, we see that

$$\langle (p_1)^{r_1-s_1}\rangle \times \langle (p_2)^{r_2-s_2}\rangle \times \cdots \times \langle (p_n)^{r_n-s_n}\rangle$$

is the required subgroup of order m. ◆

11.17 Theorem If m is a square free integer, that is, m is not divisible by the square of any prime, then every abelian group of order m is cyclic.

Proof Let G be an abelian group of square free order m. Then by Theorem 11.12, G is isomorphic to

$$\mathbb{Z}_{(p_1)^{r_1}} \times \mathbb{Z}_{(p_2)^{r_2}} \times \cdots \times \mathbb{Z}_{(p_n)^{r_n}},$$

where $m = (p_1)^{r_1}(p_2)^{r_2}\cdots(p_n)^{r_n}$. Since m is square free, we must have all $r_i = 1$ and all p_i distinct primes. Corollary 11.6 then shows that G is isomorphic to $\mathbb{Z}_{p_1p_2\cdots p_n}$, so G is cyclic. ◆

EXERCISES 11

Computations

1. List the elements of $\mathbb{Z}_2 \times \mathbb{Z}_4$. Find the order of each of the elements. Is this group cyclic?

2. Repeat Exercise 1 for the group $\mathbb{Z}_3 \times \mathbb{Z}_4$.

In Exercises 3 through 7, find the order of the given element of the direct product.

3. (2, 6) in $\mathbb{Z}_4 \times \mathbb{Z}_{12}$

4. (2, 3) in $\mathbb{Z}_6 \times \mathbb{Z}_{15}$

5. (8, 10) in $\mathbb{Z}_{12} \times \mathbb{Z}_{18}$

6. (3, 10, 9) in $\mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{15}$

7. (3, 6, 12, 16) in $\mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{20} \times \mathbb{Z}_{24}$

8. What is the largest order among the orders of all the cyclic subgroups of $\mathbb{Z}_6 \times \mathbb{Z}_8$? of $\mathbb{Z}_{12} \times \mathbb{Z}_{15}$?

9. Find all proper nontrivial subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_2$.

10. Find all proper nontrivial subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$.

11. Find all subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_4$ of order 4.

12. Find all subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_4$ that are isomorphic to the Klein 4-group.

13. Disregarding the order of the factors, write direct products of two or more groups of the form $\mathbb{Z}_n$ so that the resulting product is isomorphic to $\mathbb{Z}_{60}$ in as many ways as possible.

14. Fill in the blanks.

a. The cyclic subgroup of $\mathbb{Z}_{24}$ generated by 18 has order___.

b. $\mathbb{Z}_3 \times \mathbb{Z}_4$ is of order___.

c. The element (4, 2) of $\mathbb{Z}_{12} \times \mathbb{Z}_8$ has order___.

d. The Klein 4-group is isomorphic to $\mathbb{Z}$___ $\times$ $\mathbb{Z}$___.

e. $\mathbb{Z}_2 \times \mathbb{Z} \times \mathbb{Z}_4$ has___elements of finite order.

15. Find the maximum possible order for some element of $\mathbb{Z}_4 \times \mathbb{Z}_6$.

16. Are the groups $\mathbb{Z}_2 \times \mathbb{Z}_{12}$ and $\mathbb{Z}_4 \times \mathbb{Z}_6$ isomorphic? Why or why not?

17. Find the maximum possible order for some element of $\mathbb{Z}_8 \times \mathbb{Z}_{10} \times \mathbb{Z}_{24}$.

18. Are the groups $\mathbb{Z}_8 \times \mathbb{Z}_{10} \times \mathbb{Z}_{24}$ and $\mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{40}$ isomorphic? Why or why not?

19. Find the maximum possible order for some element of $\mathbb{Z}_4 \times \mathbb{Z}_{18} \times \mathbb{Z}_{15}$.

20. Are the groups $\mathbb{Z}_4 \times \mathbb{Z}_{18} \times \mathbb{Z}_{15}$ and $\mathbb{Z}_3 \times \mathbb{Z}_{36} \times \mathbb{Z}_{10}$ isomorphic? Why or why not?

In Exercises 21 through 25, proceed as in Example 11.13 to find all abelian groups, up to isomorphism, of the given order.

21. Order 8

22. Order 16

23. Order 32

24. Order 720

25. Order 1089

26. How many abelian groups (up to isomorphism) are there of order 24? of order 25? of order (24)(25)?

27. Following the idea suggested in Exercise 26, let m and n be relatively prime positive integers. Show that if there are (up to isomorphism) r abelian groups of order m and s of order n, then there are (up to isomorphism) rs abelian groups of order mn.

28. Use Exercise 27 to determine the number of abelian groups (up to isomorphism) of order $(10)^5$.

29. **a.** Let p be a prime number. Fill in the second row of the table to give the number of abelian groups of order p^n, up to isomorphism.

n	2	3	4	5	6	7	8
number of groups							

b. Let p, q, and r be distinct prime numbers. Use the table you created to find the number of abelian groups, up to isomorphism, of the given order.

i. $p^3q^4r^7$ **ii.** $(qr)^7$ **iii.** $q^5r^4q^3$

30. Indicate schematically a Cayley digraph for $\mathbb{Z}_m \times \mathbb{Z}_n$ for the generating set $S = \{(1, 0), (0, 1)\}$.

31. Consider Cayley digraphs with two arc types, a solid one with an arrow and a dashed one with no arrow, and consisting of two regular n-gons, for $n \geq 3$, with solid arc sides, one inside the other, with dashed arcs joining the vertices of the outer n-gon to the inner one. Figure 7.9(b) shows such a Cayley digraph with $n = 3$, and Figure 7.11(b) shows one with $n = 4$. The arrows on the outer n-gon may have the same (clockwise or counterclockwise) direction as those on the inner n-gon, or they may have the opposite direction. Let G be a group with such a Cayley digraph.

a. Under what circumstances will G be abelian?

b. If G is abelian, to what familiar group is it isomorphic?

c. If G is abelian, under what circumstances is it cyclic?

d. If G is not abelian, to what group we have discussed is it isomorphic?

Concepts

32. Mark each of the following true or false.

_____ **a.** If G_1 and G_2 are any groups, then $G_1 \times G_2$ is always isomorphic to $G_2 \times G_1$.

_____ **b.** Computation in an external direct product of groups is easy if you know how to compute in each component group.

_____ **c.** Groups of finite order must be used to form an external direct product.

_____ **d.** A group of prime order could not be the internal direct product of two proper nontrivial subgroups.

_____ **e.** $\mathbb{Z}_2 \times \mathbb{Z}_4$ is isomorphic to $\mathbb{Z}_8$.

_____ **f.** $\mathbb{Z}_2 \times \mathbb{Z}_4$ is isomorphic to S_8.

_____ **g.** $\mathbb{Z}_3 \times \mathbb{Z}_8$ is isomorphic to S_4.

_____ **h.** Every element in $\mathbb{Z}_4 \times \mathbb{Z}_8$ has order 8.

_____ **i.** The order of $\mathbb{Z}_{12} \times \mathbb{Z}_{15}$ is 60.

_____ **j.** $\mathbb{Z}_m \times \mathbb{Z}_n$ has mn elements whether m and n are relatively prime or not.

33. Give an example illustrating that not every nontrivial abelian group is the internal direct product of two proper nontrivial subgroups.

34. a. How many subgroups of $\mathbb{Z}_5 \times \mathbb{Z}_6$ are isomorphic to $\mathbb{Z}_5 \times \mathbb{Z}_6$?

b. How many subgroups of $\mathbb{Z} \times \mathbb{Z}$ are isomorphic to $\mathbb{Z} \times \mathbb{Z}$?

35. Give an example of a nontrivial group that is not of prime order and is not the internal direct product of two nontrivial subgroups.

36. Mark each of the following true or false.

_____ **a.** Every abelian group of prime order is cyclic.

_____ **b.** Every abelian group of prime power order is cyclic.

_____ **c.** $\mathbb{Z}_8$ is generated by $\{4, 6\}$.

_____ **d.** $\mathbb{Z}_8$ is generated by $\{4, 5, 6\}$.

_____ **e.** All finite abelian groups are classified up to isomorphism by Theorem 11.12.

_____ **f.** Any two finitely generated abelian groups with the same Betti number are isomorphic.

_____ **g.** Every abelian group of order divisible by 5 contains a cyclic subgroup of order 5.

_____ **h.** Every abelian group of order divisible by 4 contains a cyclic subgroup of order 4.

_____ **i.** Every abelian group of order divisible by 6 contains a cyclic subgroup of order 6.

_____ **j.** Every finite abelian group has a Betti number of 0.

37. Let p and q be distinct prime numbers. How does the number (up to isomorphism) of abelian groups of order p^r compare with the number (up to isomorphism) of abelian groups of order q^r?

38. Let G be an abelian group of order 72.

a. Can you say how many subgroups of order 8 G has? Why, or why not?

b. Can you say how many subgroups of order 4 G has? Why, or why not?

39. Let G be an abelian group. Show that the elements of finite order in G form a subgroup. This subgroup is called the **torsion subgroup** of G.

Exercises 40 through 43 deal with the concept of the torsion subgroup just defined.

40. Find the order of the torsion subgroup of $\mathbb{Z}_4 \times \mathbb{Z} \times \mathbb{Z}_3$; of $\mathbb{Z}_{12} \times \mathbb{Z} \times \mathbb{Z}_{12}$.

41. Find the torsion subgroup of the multiplicative group $\mathbb{R}^*$ of nonzero real numbers.

42. Find the torsion subgroup T of the multiplicative group $\mathbb{C}^*$ of nonzero complex numbers.

43. An abelian group is **torsion free** if e is the only element of finite order. Use Theorem 11.12 to show that every finitely generated abelian group is the internal direct product of its torsion subgroup and of a torsion-free subgroup. (Note that $\{e\}$ may be the torsion subgroup, and is also torsion free.)

44. The part of the decomposition of G in Theorem 11.12 corresponding to the subgroups of prime-power order can also be written in the form $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_r}$, where m_i divides m_{i+1} for $i = 1, 2, \cdots, r-1$. The numbers m_i can be shown to be unique, and are the **torsion coefficients** of G.

a. Find the torsion coefficients of $\mathbb{Z}_4 \times \mathbb{Z}_9$.

b. Find the torsion coefficients of $\mathbb{Z}_6 \times \mathbb{Z}_{12} \times \mathbb{Z}_{20}$.

c. Describe an algorithm to find the torsion coefficients of a direct product of cyclic groups.

Proof Synopsis

45. Give a two-sentence synopsis of the proof of Theorem 11.5.

Theory

46. Prove that a direct product of abelian groups is abelian.

47. Let G be an abelian group. Let H be the subset of G consisting of the identity e together with all elements of G of order 2. Show that H is a subgroup of G.

48. Following up the idea of Exercise 47 determine whether H will always be a subgroup for every abelian group G if H consists of the identity e together with all elements of G of order 3; of order 4. For what positive integers n will H always be a subgroup for every abelian group G, if H consists of the identity e together with all elements of G of order n? Compare with Exercise 48 of Section 5.

49. Find a counterexample of Exercise 47 with the hypothesis that G is abelian omitted.

Let H and K be subgroups of a group G. Exercises 50 and 51 ask you to establish necessary and sufficient criteria for G to appear as the internal direct product of H and K.

50. Let H and K be groups and let $G = H \times K$. Recall that both H and K appear as subgroups of G in a natural way. Show that these subgroups H (actually $H \times \{e\}$) and K (actually $\{e\} \times K$) have the following properties.

a. Every element of G is of the form hk for some $h \in H$ and $k \in K$.

b. $hk = kh$ for all $h \in H$ and $k \in K$.

c. $H \cap K = \{e\}$.

51. Let H and K be subgroups of a group G satisfying the three properties listed in the preceding exercise. Show that for each $g \in G$, the expression $g = hk$ for $h \in H$ and $k \in K$ is unique. Then let each g be renamed (h, k). Show that, under this renaming, G becomes structurally identical (isomorphic) to $H \times K$.

52. Show that a finite abelian group is not cyclic if and only if it contains a subgroup isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$ for some prime p.

53. Prove that if a finite abelian group has order a power of a prime p, then the order of every element in the group is a power of p. Can the hypothesis of commutativity be dropped? Why, or why not?

54. Let G, H, and K be finitely generated abelian groups. Show that if $G \times K$ is isomorphic to $H \times K$, then $G \simeq H$.

SECTION 12 †PLANE ISOMETRIES

Consider the Euclidean plane $\mathbb{R}^2$. An **isometry of** $\mathbb{R}^2$ is a permutation $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ that preserves distance, so that the distance between points P and Q is the same as the distance between the points $\phi(P)$ and $\phi(Q)$ for all points P and Q in $\mathbb{R}^2$. If ψ is also an isometry of $\mathbb{R}^2$, then the distance between $\psi(\phi(P))$ and $\psi(\phi(Q))$ must be the same as the distance between $\phi(P)$ and $\phi(Q)$, which in turn is the distance between P and Q, showing that the composition of two isometries is again an isometry. Since the identity map is an isometry and the inverse of an isometry is an isometry, we see that the isometries of $\mathbb{R}^2$ form a subgroup of the group of all permutations of $\mathbb{R}^2$.

Given any subset S of $\mathbb{R}^2$, the isometries of $\mathbb{R}^2$ that carry S onto itself form a subgroup of the group of isometries. This subgroup is the **group of symmetries of** S **in** $\mathbb{R}^2$. In Section 8 we gave tables for the group of symmetries of an equilateral triangle and for the group of symmetries of a square in $\mathbb{R}^2$.

Everything we have defined in the two preceding paragraphs could equally well have been done for n-dimensional Euclidean space $\mathbb{R}^n$, but we will concern ourselves chiefly with plane isometries here.

It can be proved that every isometry of the plane is one of just four types (see Artin [5]). We will list the types and show, for each type, a labeled figure that can be carried into itself by an isometry of that type. In each of Figs. 12.1, 12.3, and 12.4, consider the line with spikes shown to be extended infinitely to the left and to the right. We also give an example of each type in terms of coordinates.

translation τ: Slide every point the same distance in the same direction. See Fig. 12.1. (*Example:* $\tau(x, y) = (x, y) + (2, -3) = (x + 2, y - 3)$.)

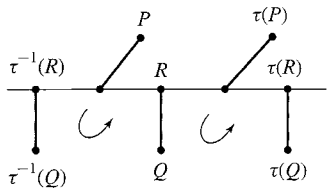
rotation ρ: Rotate the plane about a point P through an angle θ. See Fig. 12.2. (*Example:* $\rho(x, y) = (-y, x)$ is a rotation through 90° counterclockwise about the origin (0, 0).)

reflection μ: Map each point into its mirror image (μ for mirror) across a line L, each point of which is left fixed by μ. See Fig. 12.3. The line L is the *axis of reflection*. (*Example:* $\mu(x, y) = (y, x)$ is a reflection across the line $y = x$.)

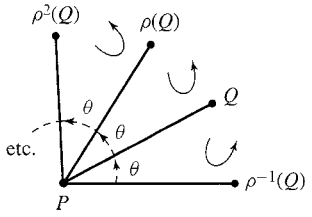
glide reflection γ: The product of a translation and a reflection across a line mapped into itself by the translation. See Fig. 12.4. (*Example:* $\gamma(x, y) = (x + 4, -y)$ is a glide reflection along the x-axis.)

Notice the little curved arrow that is carried into another curved arrow in each of Figs. 12.1 through 12.4. For the translation and rotation, the counterclockwise directions of the curved arrows remain the same, but for the reflection and glide reflection, the counterclockwise arrow is mapped into a clockwise arrow. We say that translations and rotations *preserve orientation,* while the reflection and glide reflection *reverse orientation.* We do not classify the identity isometry as any definite one of the four types listed; it could equally well be considered to be a translation by the zero vector or a rotation about any point through an angle of 0°. We always consider a glide reflection to be the product of a reflection and a translation that is different from the identity isometry.

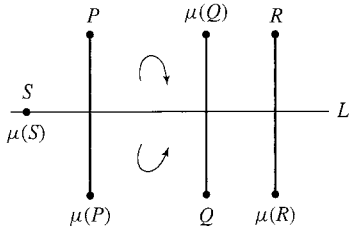
† This section is not used in the remainder of the text.



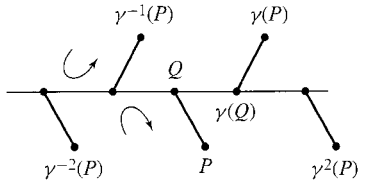
12.1 Figure Translation τ.



12.2 Figure Rotation ρ.



12.3 Figure Reflection μ.



12.4 Figure Glide reflection γ.

The theorem that follows describes the possible structures of finite subgroups of the full isometry group.

12.5 Theorem Every finite group G of isometries of the plane is isomorphic to either $\mathbb{Z}_n$ or to a dihedral group D_n for some positive integer n.

Proof Outline First we show that there is a point P in the plane that is left fixed by every isometry in G. This can be done in the following way, using coordinates in the plane. Suppose $G = \{\phi_1, \phi_2, \cdots, \phi_m\}$ and let

$$(x_i, y_i) = \phi_i(0, 0).$$

Then the point

$$P = (\overline{x}, \overline{y}) = \left(\frac{x_1 + x_2 + \cdots + x_m}{m}, \frac{y_1 + y_2 + \cdots + y_m}{m} \right)$$

is the *centroid* of the set $S = \{(x_i, y_i) \mid i = 1, 2, \cdots, m\}$. The isometries in G permute the points in S among themselves, since if $\phi_i\phi_j = \phi_k$ then $\phi_i(x_j, y_j) = \phi_i\{\phi_j(0, 0)] = \phi_k(0, 0) = (x_k, y_k)$. It can be shown that the centroid of a set of points is uniquely determined by its distances from the points, and since each isometry in G just permutes the set S, it must leave the centroid $(\overline{x}, \overline{y})$ fixed. Thus G consists of the identity, rotations about P, and reflections across a line through P.

The orientation-preserving isometries in G form a subgroup H of G which is either all of G or of order $m/2$. This can be shown in the same way that we showed that the even permutations are a subgroup of S_n containing just half the elements of S_n. (See Exercise 22.) Of course H consists of the identity and the rotations in G. If we choose a rotation in G that rotates the plane through as small an angle $\theta > 0$ as possible, it can be shown to generate the subgroup H. (See Exercise 23.) This shows that if $H = G$, then G is cyclic of order m and thus isomorphic to $\mathbb{Z}_m$. Suppose $H \neq G$ so that G contains

some reflections. Let $H = \{\iota, \rho_1, \cdots, \rho_{n-1}\}$ with $n = m/2$. If μ is a reflection in G, then the coset $H\mu$ consists of all n of the reflections in G.

Consider now a regular n-gon in the plane having P as its center and with a vertex lying on the line through P left fixed by μ. The elements of H rotate this n-gon through all positions, and the elements of $H\mu$ first reflect in an axis through a vertex, effectively turning the n-gon over, and then rotate through all positions. Thus the action of G on this n-gon is the action of D_n, so G is isomorphic to D_n. ◆

The preceding theorem gives the complete story about finite plane isometry groups. We turn now to some infinite groups of plane isometries that arise naturally in decorating and art. Among these are the *discrete frieze groups*. A discrete frieze consists of a pattern of finite width and height that is repeated endlessly in both directions along its baseline to form a strip of infinite length but finite height; think of it as a decorative border strip that goes around a room next to the ceiling on wallpaper. We consider those isometries that carry each basic pattern onto itself or onto another instance of the pattern in the frieze. The set of all such isometries is called the **"frieze group."** All discrete frieze groups are infinite and have a subgroup isomorphic to $\mathbb{Z}$ generated by the translation that slides the frieze lengthwise until the basic pattern is superimposed on the position of its next neighbor pattern in that direction. As a simple example of a discrete frieze, consider integral signs spaced equal distances apart and continuing infinitely to the left and right, indicated schematically like this.

$$\cdots\int\int\int

the horizontal line through the centers of the D's, but the translation component of each glide reflection is also in the group so they are all considered trivial glide reflections in that group. The frieze group for

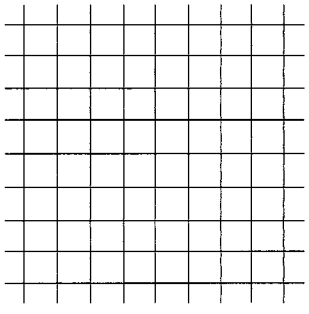
$$\cdots \quad \mathbf{D} \qquad \mathbf{D} \qquad \mathbf{D} \qquad \mathbf{D} \qquad \mathbf{D} \quad \cdots$$
$$\cdots \quad \mathbf{D} \qquad \mathbf{D} \qquad \mathbf{D} \qquad \mathbf{D} \qquad \mathbf{D} \quad \cdots$$

contains a nontrivial glide reflection whose translation component is not an element of the group. The exercises exhibit the seven possible cases, and ask you to tell, for each case, which of the four types of isometries displayed above appear in the symmetry group. We do not obtain seven different group structures. Each of the groups obtained can be shown to be isomorphic to one of

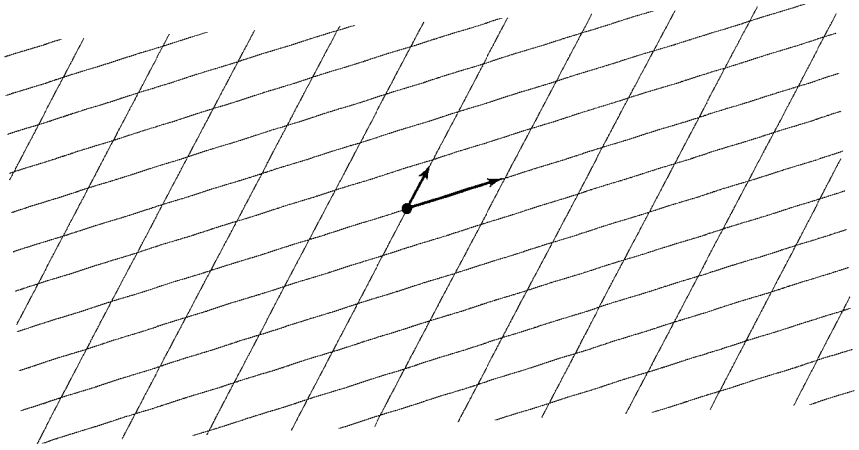
$$\mathbb{Z}, \quad D_\infty, \quad \mathbb{Z} \times \mathbb{Z}_2, \quad \text{or} \quad D_\infty \times \mathbb{Z}_2.$$

Equally interesting is the study of symmetries when a pattern in the shape of a square, parallelogram, rhombus, or hexagon is repeated by translations along *two nonparallel vector directions* to fill the entire plane, like patterns that appear on wallpaper. These groups are called the *wallpaper groups* or the *plane crystallographic groups*. While a frieze could not be carried into itself by a rotation through a positive angle less than 180°, it is possible to have rotations of 60°, 90°, 120°, and 180° for some of these plane-filling patterns. Figure 12.6 provides an illustration where the pattern consists of a square. We are interested in the group of plane isometries that carry this square onto itself or onto another square. Generators for this group are given by two translations (one sliding a square to the next neighbor to the right and one to the next above), by a rotation through 90° about the center of a square, and by a reflection in a vertical (or horizontal) line along the edges of the square. The one reflection is all that is needed to "turn the plane over"; a diagonal reflection can also be used. After being turned over, the translations and rotations can be used again. The isometry group for this *periodic pattern* in the plane surely contains a subgroup isomorphic to $\mathbb{Z} \times \mathbb{Z}$ generated by the unit translations to the right and upward, and a subgroup isomorphic to D_4 generated by those isometries that carry one square (it can be any square) into itself.

If we consider the plane to be filled with parallelograms as in Fig. 12.7, we do not get all the types of isometries that we did for Fig. 12.6. The symmetry group this time is



12.6 Figure



12.7 Figure

generated by the translations indicated by the arrows and a rotation through 180° about any vertex of a parallelogram.

It can be shown that there are 17 different types of wallpaper patterns when they are classified according to the types of rotations, reflections, and nontrivial glide reflections that they admit. We refer you to Gallian [8] for pictures of these 17 possibilities and a chart to help you identify them. The exercises illustrate a few of them. The situation in space is more complicated; it can be shown that there are 230 three-dimensional crystallographic groups. The final exercise we give involves rotations in space.

M. C. Escher (1898–1973) was an artist whose work included plane-filling patterns. The exercises include reproductions of four of his works of this type.

EXERCISES 12

1. This exercise shows that the group of symmetries of a certain type of geometric figure may depend on the dimension of the space in which we consider the figure to lie.

- **a.** Describe all symmetries of a point in the real line $\mathbb{R}$; that is, describe all isometries of $\mathbb{R}$ that leave one point fixed.
- **b.** Describe all symmetries (translations, reflections, etc.) of a point in the plane $\mathbb{R}^2$.
- **c.** Describe all symmetries of a line segment in $\mathbb{R}$.
- **d.** Describe all symmetries of a line segment in $\mathbb{R}^2$.
- **e.** Describe some symmetries of a line segment in $\mathbb{R}^3$.

2. Let P stand for an orientation preserving plane isometry and R for an orientation reversing one. Fill in the table with P or R to denote the orientation preserving or reversing property of a product.

	P	R
P		
R		

3. Fill in the table to give *all* possible types of plane isometries given by a product of two types. For example, a product of two rotations may be a rotation, or it may be another type. Fill in the box corresponding to $\rho\rho$ with both letters. Use your answer to Exercise 2 to eliminate some types. Eliminate the identity from consideration.

	τ	ρ	μ	γ
τ				
ρ				
μ				
γ				

4. Draw a plane figure that has a one-element group as its group of symmetries in $\mathbb{R}^2$.
5. Draw a plane figure that has a two-element group as its group of symmetries in $\mathbb{R}^2$.
6. Draw a plane figure that has a three-element group as its group of symmetries in $\mathbb{R}^2$.
7. Draw a plane figure that has a four-element group isomorphic to $\mathbb{Z}_4$ as its group of symmetries in $\mathbb{R}^2$.
8. Draw a plane figure that has a four-element group isomorphic to the Klein 4-group V as its group of symmetries in $\mathbb{R}^2$.
9. For each of the four types of plane isometries (other than the identity), give the possibilities for the order of an isometry of that type in the group of plane isometries.
10. A plane isometry ϕ has a *fixed point* if there exists a point P in the plane such that $\phi(P) = P$. Which of the four types of plane isometries (other than the identity) can have a fixed point?
11. Referring to Exercise 10, which types of plane isometries, if any, have exactly one fixed point?
12. Referring to Exercise 10, which types of plane isometries, if any, have exactly two fixed points?
13. Referring to Exercise 10, which types of plane isometries, if any, have an infinite number of fixed points?
14. Argue geometrically that a plane isometry that leaves three noncolinear points fixed must be the identity map.
15. Using Exercise 14, show algebraically that if two plane isometries ϕ and ψ agree on three noncolinear points, that is, if $\phi(P_i) = \psi(P_i)$ for noncolinear points P_1, P_2, and P_3, then ϕ and ψ are the same map.
16. Do the rotations, together with the identity map, form a subgroup of the group of plane isometries? Why or why not?
17. Do the translations, together with the identity map, form a subgroup of the group of plane isometries? Why or why not?
18. Do the rotations about one particular point P, together with the identity map, form a subgroup of the group of plane isometries? Why or why not?
19. Does the reflection across one particular line L, together with the identity map, form a subgroup of the group of plane isometries? Why or why not?
20. Do the glide reflections, together with the identity map, form a subgroup of the group of plane isometries? Why or why not?
21. Which of the four types of plane isometries can be elements of a *finite* subgroup of the group of plane isometries?
22. Completing a detail of the proof of Theorem 12.5, let G be a finite group of plane isometries. Show that the rotations in G, together with the identity isometry, form a subgroup H of G, and that either $H = G$ or $|G| = 2|H|$. [*Hint:* Use the same method that we used to show that $|S_n| = 2|A_n|$.]

23. Completing a detail in the proof of Theorem 12.5, let G be a finite group consisting of the identity isometry and rotations about one point P in the plane. Show that G is cyclic, generated by the rotation in G that turns the plane counterclockwise about P through the smallest angle $\theta > 0$. [*Hint:* Follow the idea of the proof that a subgroup of a cyclic group is cyclic.]

Exercises 24 through 30 illustrate the seven different types of friezes when they are classified according to their symmetries. Imagine the figure shown to be continued infinitely to the right and left. The symmetry group of a frieze always contains translations. For each of these exercises answer these questions about the symmetry group of the frieze.

a. Does the group contain a rotation?

b. Does the group contain a reflection across a horizontal line?

c. Does the group contain a reflection across a vertical line?

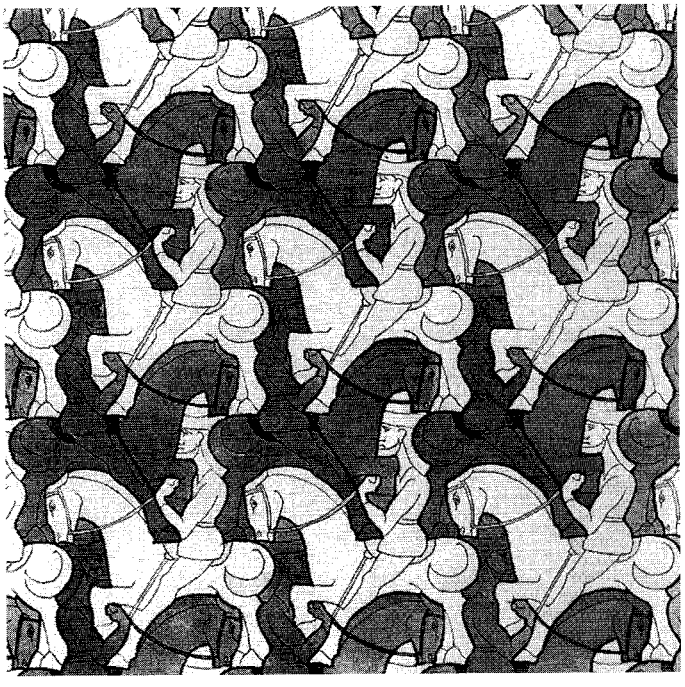
d. Does the group contain a nontrivial glide reflection?

e. To which of the possible groups $\mathbb{Z}$, D_∞, $\mathbb{Z} \times \mathbb{Z}_2$, or $D_\infty \times \mathbb{Z}_2$ do you think the symmetry group of the frieze is isomorphic?

24. F F F F F F F F F F F F F F

25. T T T T T T T T T T

26. E E E E E E E E E E E E



12.8 Figure The Study of Regular Division of the Plane with Horsemen (© 1946 M. C. Escher Foundation–Baarn–Holland. All rights reserved.)

27. Z Z Z Z Z Z Z Z Z Z Z Z

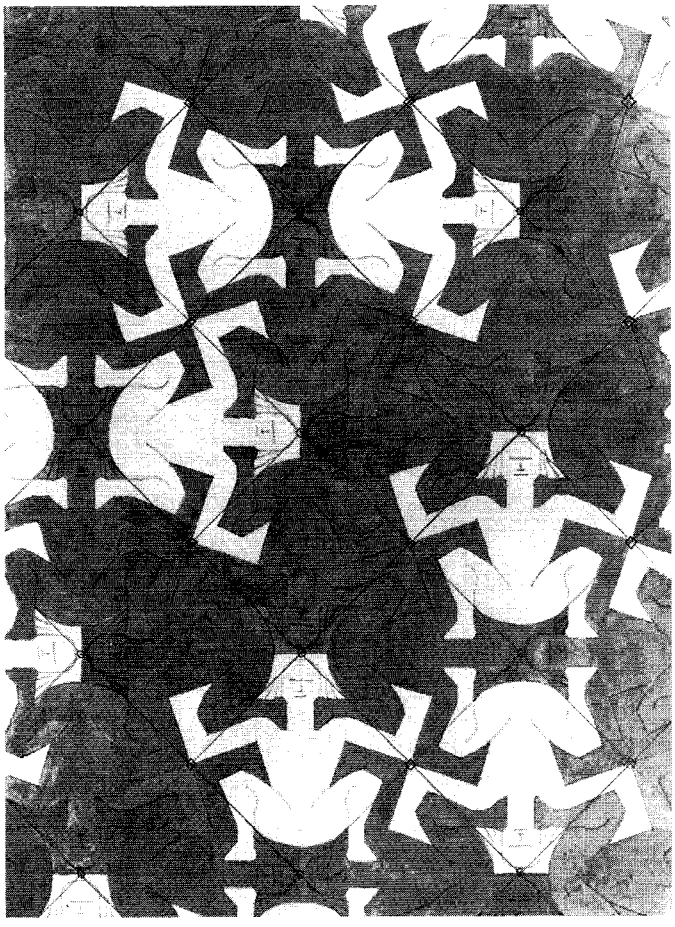
28. H H H H H H H H H H H H

29.

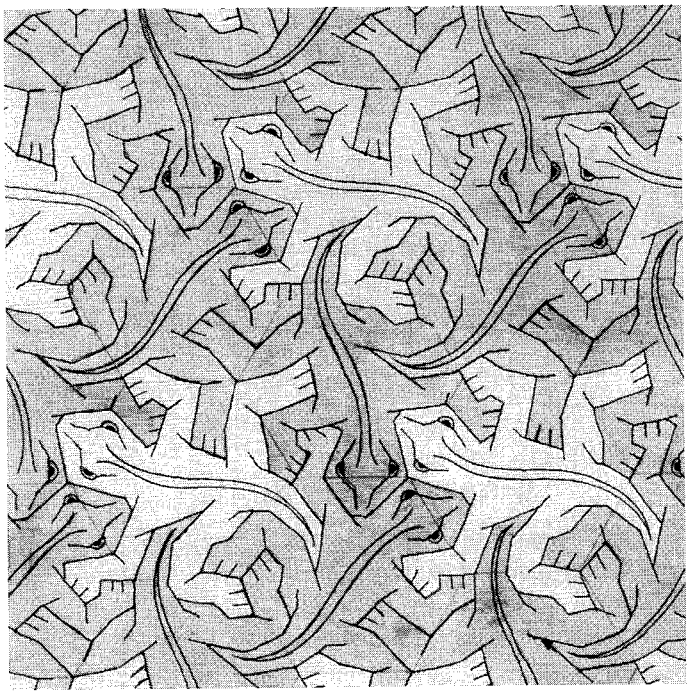
30.

Exercises 31 through 37 describe a pattern to be used to fill the plane by translation in the two directions given by the specified vectors. Answer these questions in each case.

a. Does the symmetry group contain any rotations? If so, through what possible angles θ where $0 < \theta \leq 180°$?



12.9 Figure The Study of Regular Division of the Plane with Imaginary Human Figures (© 1936 M. C. Escher Foundation–Baarn–Holland. All rights reserved.)



12.10 Figure The Study of Regular Division of the Plane with Reptiles (© 1939 M. C. Escher Foundation–Baarn–Holland. All rights reserved.)

b. Does the symmetry group contain any reflections?

c. Does the symmetry group contain any nontrivial glide reflections?

31. A square with horizontal and vertical edges using translation directions given by vectors (1, 0) and (0, 1).

32. A square as in Exercise 31 using translation directions given by vectors (1, 1/2) and (0, 1).

33. A square as in Exercise 31 with the letter L at its center using translation directions given by vectors (1, 0) and (0, 1).

34. A square as in Exercise 31 with the letter E at its center using translation directions given by vectors (1, 0) and (0, 1).

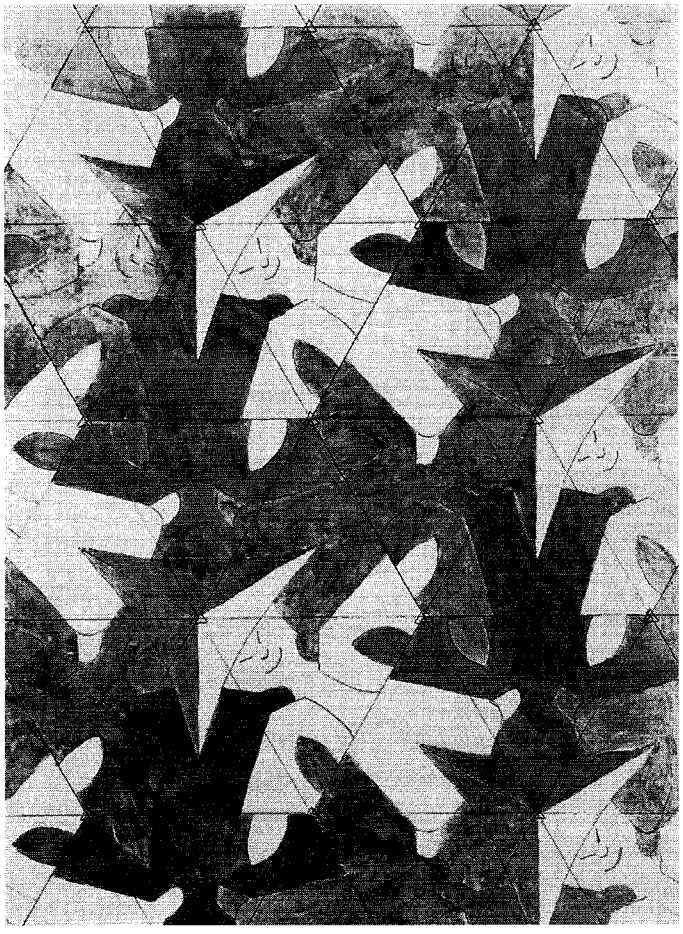
35. A square as in Exercise 31 with the letter H at its center using translation directions given by vectors (1, 0) and (0, 1).

36. A regular hexagon with a vertex at the top using translation directions given by vectors (1, 0) and $(1, \sqrt{3})$.

37. A regular hexagon with a vertex at the top containing an equilateral triangle with vertex at the top and centroid at the center of the hexagon, using translation directions given by vectors (1, 0) and $(1, \sqrt{3})$.

Exercises 38 through 41 are concerned with art works of M. C. Escher. Neglect the shading in the figures and assume the markings in each human figure, reptile, or horseman are the same, even though they may be invisible due to shading. Answer the same questions (a), (b), and (c) that were asked for Exercises 31 through 36, and also answer this part (d).

d. Assuming horizontal and vertical coordinate axes with equal scales as usual, give vectors in the two nonparallel directions of vectors that generate the translation subgroup. Do not concern yourself with the length of these vectors.



12.11 Figure The Study of Regular Division of the Plane with Human Figures (© 1936 M. C. Escher Foundation–Baarn–Holland. All rights reserved.)

38. *The Study of Regular Division of the Plane with Horsemen* in Fig. 12.8.

39. *The Study of Regular Division of the Plane with Imaginary Human Figures* in Fig. 12.9.

40. *The Study of Regular Division of the Plane with Reptiles* in Fig. 12.10.

41. *The Study of Regular Division of the Plane with Human Figures* in Fig. 12.11.

42. Show that the rotations of a cube in space form a group isomorphic to S_4. [*Hint:* A rotation of the cube permutes the diagonals through the center of the cube.]